

my map

run my map JCL

vsam_SKILLFILE

```

/* STEP 1: DEFINE VSAM KSDS FILE *
/*****
//STEP1    EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSIN    DD *
    DELETE TSOCP01.ESK0G.SKILLS CLUSTER PURGE
    IF LASTCC <= 8 THEN SET MAXCC = 0

    DEFINE CLUSTER -
      (NAME(TSOCP01.ESK0G.SKILLS) -
      CYLINDERS(1 1) -
      INDEXED -
      KEYS(4 0) -
      RECORDSIZE(240 240) -
      FREESPACE(10 10) -
      SHAREOPTIONS(2 3) -
      VOLUMES(SYSDA)) -
    DATA -
      (NAME(TSOCP01.ESK0G.SKILLS.DATA)) -
    INDEX -
      (NAME(TSOCP01.ESK0G.SKILLS.INDEX))
/*

```

```
//STEP1    EXEC PGM=IDCAMS
```

```
//SYSPRINT DD SYSOUT=*
```

```
//SYSIN    DD *
```

```
DELETE TSOCP01.ESK0G.SKILLS CLUSTER PURGE
```

```
IF LASTCC <= 8 THEN SET MAXCC = 0
```

```
DEFINE CLUSTER -
```

```
(NAME(TSOCP01.ESK0G.SKILLS) -
```

```
CYLINDERS(1 1) -
```

INDEXED -

KEYS(4 0) -

RECORDSIZE(240 240) -

FREESPACE(10 10) -

SHAREOPTIONS(2 3) -

VOLUMES(SYSDA)) -

DATA -

(NAME(TSOCP01.ESK0G.SKILLS.DATA)) -

INDEX -

(NAME(TSOCP01.ESK0G.SKILLS.INDEX))

/*

Job 1960 - cc 0012

FIX: Shorten **SKILLFILE to an 8-character name like **SKILLFIL** or **SKILLS** in your JCL.**

```

IDCAMS  SYSTEM SERVICES                                TIME: 05:35:
0
  DELETE TS0CP01.ESK0G.SKILLFILE CLUSTER PURGE
0IDC3203I ITEM 'TS0CP01.ESK0G.SKILLF' DOES NOT ADHERE TO RESTRICTIONS
0IDC3202I ABOVE TEXT BYPASSED UNTIL NEXT COMMAND. CONDITION CODE IS 12
0
  SET LASTCC = 0

  DEFINE CLUSTER -
    (NAME(TS0CP01.ESK0G.SKILLFILE) -
0IDC3203I ITEM 'TS0CP01.ESK0G.SKILLF' DOES NOT ADHERE TO RESTRICTIONS
0    INDEXED -
      KEYS(4 0) -
      RECORDSIZE(240 240) -
      FREESPACE(10 10) -
      SHAREOPTIONS(2 3) -
      VOLUMES(SYSDA)) -
    DATA -
      (NAME(TS0CP01.ESK0G.SKILLFILE.DATA)) -
    INDEX -
      (NAME(TS0CP01.ESK0G.SKILLFILE.INDEX))
0IDC3202I ABOVE TEXT BYPASSED UNTIL NEXT COMMAND. CONDITION CODE IS 12
0
0IDC0002I IDCAMS PROCESSING COMPLETE. MAXIMUM CONDITION CODE WAS 12

```

**//DEFINEJOB JOB (ACCOUNT),'DEFINE
VSAM',CLASS=A,MSGCLASS=X,NOTIFY=&SYSUID**

//*****

**/* DEFINE VSAM KSDS FILE FOR SKILLS & DETAILS
(RECORD LENGTH 240) ***


```
//*****  
*
```

```
//STEP1 EXEC PGM=IDCAMS
```

```
//SYSPRINT DD SYSOUT=*
```

```
//SYSIN DD *
```

```
DELETE YOUR.HLQ.SKILLFILE CLUSTER PURGE
```

```
SET LASTCC = 0
```

```
DEFINE CLUSTER -
```

```
(NAME(YOUR.HLQ.SKILLFILE) -
```

```
INDEXED -
```

```
KEYS(4 0) -
```

```
RECORDSIZE(240 240) -
```

```
FREESPACE(10 10) -
```

```
SHAREOPTIONS(2 3) -
```

```
VOLUMES(SYSDA)) -
```

```
DATA -
```

```
(NAME(YOUR.HLQ.SKILLFILE.DATA)) -
```

INDEX -

(NAME(YOUR.HLQ.SKILLFILE.INDEX))

/*

1. Store it in a PDS instead of VSAM (Recommended)

Allocate a standard Partitioned Data Set using IDCAMS or Zowe Explorer with these parameters:

- **Data Set Name:** YOUR.PREFIX.BMS(MFRNMAP)
- **Record Format (RECFM):** FB (Fixed Block)
- **Record Length (LRECL):** 80
- **Block Size (BLKSIZE):** 27920 (or 6160)

Field Name	Description	BMS Length
MFYEAR	Year input	4 bytes
MFCAT	Category code (S/P)	1 byte

MFTITLE	Role / Title	40 bytes
MFSKILL	Skills	40 bytes
MFHL1	Highlight line 1	40 bytes
MFHL2	Highlight line 2	40 bytes
MFHL3	Highlight line 3	40 bytes
MFLINK	Link / Reference	40 bytes
Total Payload		245 bytes

Recommended VSAM Record Design

- **Fixed-Length File: 245 bytes** (`RECORDSIZE(245 245)`).
- **Primary Key (KSDS): 5 bytes** using `MFYEAR` (4 bytes) + `MFCAT` (1 byte) defined at offset 0 (`KEYS(5 0)`).

If you want to leave room for future expansion or metadata (like a record status flag or timestamp), rounding the

record length up to a standard size like **256** or **512 bytes** with padding is common practice.

myprogram

IDENTIFICATION DIVISION.

PROGRAM-ID. MFRNMAPP.

AUTHOR. FIONA.

=====

* FULL CRUD CICS PROGRAM WITH ALL FUNCTION KEYS SUPPORTED *

=====

ENVIRONMENT DIVISION.

DATA DIVISION.

WORKING-STORAGE SECTION.

*--- BMS MAP COPYBOOK ---

COPY MFRNSET.

*--- DFHAID & DFHBMSCA COPYBOOKS ---

COPY DFHAID.

COPY DFHBMSCA.

*--- VSAM RECORD STRUCTURE ---

01 WS-SKILL-DETAIL-RECORD.

05 WS-REC-YEAR PIC X(04).

05 WS-REC-CATEGORY PIC X(01).

05 WS-REC-ROLE-TITLE PIC X(40).

05 WS-REC-SKILLS PIC X(40).

05 WS-REC-HIGHLIGHTS.

10 WS-REC-HL1 PIC X(40).

10 WS-REC-HL2 PIC X(40).

10 WS-REC-HL3 PIC X(40).

05 WS-REC-LINK-REF PIC X(40).

05 FILLER PIC X(35) VALUE SPACES.

*--- PROGRAM CONTROL FLAGS & WORK FIELDS ---

01 WS-FLAGS.

05 VALID-DATA-SW PIC X(01) VALUE 'Y'.

88 VALID-DATA VALUE 'Y'.

88 INVALID-DATA VALUE 'N'.

05 SEND-FLAG PIC X(01) VALUE 'E'.

88 SEND-ERASE VALUE 'E'.

88 SEND-DATAONLY VALUE 'D'.

88 SEND-ERROR VALUE 'R'.

05 BROWSE-FLG PIC X(01) VALUE 'N'.

88 IN-BROWSE VALUE 'Y'.

01 WS-CICS-WORK-AREAS.

05 RESPCODE PIC S9(08) COMP VALUE ZERO.

```
EVALUATE EIBALD
  WHEN DFHPF1
    PERFORM 100-READPARA
  WHEN DFHPF2
    PERFORM 200-WRITEPARA
  WHEN DFHPF3
    PERFORM 300-EXITPARA
  WHEN DFHPF4
    PERFORM 400-UPDATEPARA
  WHEN DFHPF5
    PERFORM 500-DELETEPARA
  WHEN DFHPF6
    PERFORM 600-CLEARPARA
  WHEN DFHPF7
    PERFORM 700-READPREVPARA
  WHEN DFHPF8
    PERFORM 800-READNEXTPARA
  WHEN DFHENTER
    PERFORM 000-BROWSEPARA
  WHEN DFHPF10
```

```
        PERFORM 450-CONF-UPDATE
    WHEN DFHCLEAR
        PERFORM 050-SEND-INITIAL-MAP
    WHEN OTHER
        PERFORM 110-RECEIVE-PARA
        MOVE 'Invalid Key Pressed' TO MFMSGO
        SET SEND-DATAONLY TO TRUE
        PERFORM 900-SEND-PARA
    END-EVALUATE
END-IF.
```

```
EXEC CICS RETURN
    TRANSID('MYTRN')
    COMMAREA(WC-COMM-AREA)
    LENGTH(LENGTH OF WC-COMM-AREA)
END-EXEC.
```

```
*=====*
* INITIAL MAP & UTILITY PARAGRAPHS                      *
*=====*
```

```
050-SEND-INITIAL-MAP.
    MOVE LOW-VALUES TO MFRNMAPO.
    PERFORM 050-MOVEPARA.
    MOVE 'Enter Year and press F1 to Read or F2 to Add.'
      TO MFMSGO.
    MOVE -1 TO MFYEARL.
    SET SEND-ERASE TO TRUE.
    PERFORM 900-SEND-PARA.
```

```
050-ASKTIMEPARA.
    EXEC CICS
        ASKTIME
        ABSTIME(TSTAMP)
    END-EXEC.
    EXEC CICS
        FORMATTIME
        ABSTIME(TSTAMP)
        YYYYMMDD(OUTDATE)
        DATESEP('/')
        TIME(OUTTIME)
        TIMESEP(':')
    END-EXEC.
```

```
050-MOVEPARA.
```


MOVE OUTDATE TO MFDATEO.
MOVE OUTTIME TO MFTIMEO.

600-CLEARPARA.
MOVE 'N' TO WS-COMM-BROWSE-SW.
PERFORM 050-SEND-INITIAL-MAP.

300-EXITPARA.
EXEC CICS SEND CONTROL ERASE END-EXEC.
EXEC CICS RETURN END-EXEC.

=====

* F1: READ RECORD (INQUIRY BY YEAR) *

=====

100-READPARA.
MOVE SPACES TO MFMSGO.
PERFORM 110-RECEIVE-PARA.
PERFORM 120-VALIDATE-YEAR.

IF INVALID-DATA
SET SEND-ERROR TO TRUE
MOVE -1 TO MFYEARL
PERFORM 900-SEND-PARA
GO TO 100-EXIT
END-IF.

MOVE MFYEARI TO WS-REC-YEAR.

EXEC CICS
READ FILE('SKILLFILE')
INTO(WS-SKILL-DETAIL-RECORD)
RIDFLD(WS-REC-YEAR)
RESP(RESPCODE)
END-EXEC.

IF RESP CODE = DFHRESP(NORMAL)
PERFORM 180-MOVE-REC-MAP-PARA
MOVE 'Record Found.' TO MFMSGO
SET SEND-DATAONLY TO TRUE
MOVE -1 TO MFCATL
ELSE IF RESP CODE = DFHRESP(NOTFND)
MOVE 'Record Not Found.' TO MFMSGO
SET SEND-ERROR TO TRUE
MOVE -1 TO MFYEARL

```
ELSE
  MOVE 'Error Reading VSAM File.' TO MFMSGO
  SET SEND-ERROR TO TRUE
END-IF.
PERFORM 900-SEND-PARA.
100-EXIT. EXIT.
```

```
*=====*
* F2: WRITE/ADD NEW RECORD                                *
*=====*
```

```
200-WRITEPARA.
  MOVE SPACES TO MFMSGO.
  PERFORM 110-RECEIVE-PARA.
  PERFORM 120-VALIDATE-YEAR.
  PERFORM 150-VALIDATE-INPUT.
```

```
IF INVALID-DATA
  SET SEND-ERROR TO TRUE
  MOVE -1 TO MFYEARL
  PERFORM 900-SEND-PARA
  GO TO 200-EXIT
END-IF.
```

```
INITIALIZE WS-SKILL-DETAIL-RECORD.
PERFORM 170-MOVE-MAP-REC-PARA.
```

```
EXEC CICS WRITE
  FILE('SKILLFILE')
  FROM(WS-SKILL-DETAIL-RECORD)
  RIDFLD(WS-REC-YEAR)
  RESP(RESPCODE)
END-EXEC.
```

```
IF RESPCODE = DFHRESP(NORMAL)
  MOVE 'Record Written Successfully.' TO MFMSGO
ELSE IF RESPCODE = DFHRESP(DUPREC)
  MOVE 'Record Already Exists.' TO MFMSGO
ELSE
  MOVE 'Record Write Unsuccessful.' TO MFMSGO
END-IF.
SET SEND-DATAONLY TO TRUE.
MOVE -1 TO MFYEARL.
PERFORM 900-SEND-PARA.
200-EXIT. EXIT.
```

```
*=====*
* F4 & F10: READ FOR UPDATE & CONFIRM UPDATE      *
*=====*
```

400-UPDATEPARA.

MOVE SPACES TO MFMSGO.
PERFORM 110-RECEIVE-PARA.
PERFORM 120-VALIDATE-YEAR.

IF INVALID-DATA
SET SEND-ERROR TO TRUE
MOVE -1 TO MFYEARL
PERFORM 900-SEND-PARA
GO TO 400-EXIT
END-IF.

MOVE MFYEARI TO WS-REC-YEAR.

EXEC CICS
READ FILE('SKILLFILE')
INTO(WS-SKILL-DETAIL-RECORD)
RIDFLD(WS-REC-YEAR)
RESP(RESPCODE)
END-EXEC.

IF RESP CODE = DFHRESP(NORMAL)
PERFORM 180-MOVE-REC-MAP-PARA
MOVE 'Record Ready for Update. Edit and press F10.'
TO MFMSGO
SET SEND-DATAONLY TO TRUE
MOVE -1 TO MFCATL
ELSE
MOVE 'Record Not Found for Update.' TO MFMSGO
SET SEND-ERROR TO TRUE
MOVE -1 TO MFYEARL
END-IF.
PERFORM 900-SEND-PARA.

400-EXIT. EXIT.

450-CONF-UPDATE.

PERFORM 110-RECEIVE-PARA.
PERFORM 120-VALIDATE-YEAR.

IF INVALID-DATA

SET SEND-ERROR TO TRUE
MOVE -1 TO MFYEARL
PERFORM 900-SEND-PARA
GO TO 450-EXIT
END-IF.

MOVE MFYEARI TO WS-REC-YEAR.

EXEC CICS READ
FILE('SKILLFILE')
INTO(WS-SKILL-DETAIL-RECORD)
RIDFLD(WS-REC-YEAR)
UPDATE
RESP(RESPCODE)
END-EXEC.

IF RESP CODE = DFHRESP(NORMAL)
PERFORM 170-MOVE-MAP-REC-PARA
EXEC CICS REWRITE
FILE('SKILLFILE')
FROM(WS-SKILL-DETAIL-RECORD)
RESP(RESPCODE)
END-EXEC
IF RESP CODE = DFHRESP(NORMAL)
MOVE 'Record Updated Successfully.' TO MFMSGO
ELSE
MOVE 'Update Failed on REWRITE.' TO MFMSGO
END-IF
ELSE
MOVE 'Record Lock Failed. Update Aborted.' TO MFMSGO
END-IF.

SET SEND-DATAONLY TO TRUE.
MOVE -1 TO MFYEARL.
PERFORM 900-SEND-PARA.
450-EXIT. EXIT.

```
*=====*
```

```
* F5: DELETE RECORD *
```

```
*=====*
```

500-DELETEPARA.
MOVE SPACES TO MFMSGO.
PERFORM 110-RECEIVE-PARA.
PERFORM 120-VALIDATE-YEAR.

```
IF INVALID-DATA
  SET SEND-ERROR TO TRUE
  MOVE -1 TO MFYEARL
  PERFORM 900-SEND-PARA
  GO TO 500-EXIT
END-IF.
```

```
MOVE MFYEARI TO WS-REC-YEAR.
```

```
EXEC CICS DELETE
  FILE('SKILLFILE')
  RIDFLD(WS-REC-YEAR)
  RESP(RESPCODE)
END-EXEC.
```

```
IF RESPCODE = DFHRESP(NORMAL)
  MOVE 'Record Deleted Successfully.' TO MFMSGO
ELSE IF RESPCODE = DFHRESP(NOTFND)
  MOVE 'Record Not Found for Delete.' TO MFMSGO
ELSE
  MOVE 'Delete Failed.' TO MFMSGO
END-IF.
SET SEND-DATAONLY TO TRUE.
MOVE -1 TO MFYEARL.
PERFORM 900-SEND-PARA.
500-EXIT. EXIT.
```

```
*=====*
* ENTER, F7 & F8: VSAM BROWSE OPERATIONS          *
*=====*
```

```
000-BROWSEPARA.
  PERFORM 110-RECEIVE-PARA.
  PERFORM 120-VALIDATE-YEAR.
```

```
IF INVALID-DATA
  MOVE SPACES TO WS-REC-YEAR
ELSE
  MOVE MFYEARI TO WS-REC-YEAR
END-IF.
```

```
EXEC CICS STARTBR
  FILE('SKILLFILE')
  RIDFLD(WS-REC-YEAR)
```

GTEQ
RESP(RESPCODE)
END-EXEC.

IF RESPCODE = DFHRESP(NORMAL)
EXEC CICS READNEXT
FILE('SKILLFILE')
INTO(WO-SKILL-DETAIL-RECORD)
RIDFLD(WO-REC-YEAR)
RESP(RESPCODE)
END-EXEC
IF RESPCODE = DFHRESP(NORMAL)
PERFORM 180-MOVE-REC-MAP-PARA
MOVE 'Browse Started. Press F8 (Next) or F7 (Prev).'
TO MFMSGO
MOVE 'Y' TO WO-COMM-BROWSE-SW
ELSE
MOVE 'No Records Found to Browse.' TO MFMSGO
END-IF
EXEC CICS ENDBR FILE('SKILLFILE') END-EXEC
ELSE
MOVE 'Error Starting Browse Session.' TO MFMSGO
END-IF.
SET SEND-DATAONLY TO TRUE.
PERFORM 900-SEND-PARA.

700-READPREVPARA.
PERFORM 110-RECEIVE-PARA.
MOVE MFYEAR1 TO WO-REC-YEAR.

EXEC CICS STARTBR
FILE('SKILLFILE')
RIDFLD(WO-REC-YEAR)
GTEQ
RESP(RESPCODE)
END-EXEC.

IF RESPCODE = DFHRESP(NORMAL)
EXEC CICS READPREV
FILE('SKILLFILE')
INTO(WO-SKILL-DETAIL-RECORD)
RIDFLD(WO-REC-YEAR)
RESP(RESPCODE)
END-EXEC

```
IF RESPCODE = DFHRESP(NORMAL)
  PERFORM 180-MOVE-REC-MAP-PARA
  MOVE 'Previous Record Retrieved.' TO MFMSGO
ELSE
  MOVE 'Beginning of File Reached.' TO MFMSGO
END-IF
EXEC CICS ENDBR FILE('SKILLFILE') END-EXEC
ELSE
  MOVE 'Error Navigating Backward.' TO MFMSGO
END-IF.
SET SEND-DATAONLY TO TRUE.
PERFORM 900-SEND-PARA.
```

```
800-READNEXTPARA.
PERFORM 110-RECEIVE-PARA.
MOVE MFYEARI TO WS-REC-YEAR.
```

```
EXEC CICS STARTBR
  FILE('SKILLFILE')
  RIDFLD(WS-REC-YEAR)
  GTEQ
  RESP(RESPCODE)
END-EXEC.
```

```
IF RESPCODE = DFHRESP(NORMAL)
*  -- Skip current record to advance cursor --
  EXEC CICS READNEXT
    FILE('SKILLFILE')
    INTO(WS-SKILL-DETAIL-RECORD)
    RIDFLD(WS-REC-YEAR)
    RESP(RESPCODE)
  END-EXEC
  EXEC CICS READNEXT
    FILE('SKILLFILE')
    INTO(WS-SKILL-DETAIL-RECORD)
    RIDFLD(WS-REC-YEAR)
    RESP(RESPCODE)
  END-EXEC
  IF RESPCODE = DFHRESP(NORMAL)
    PERFORM 180-MOVE-REC-MAP-PARA
    MOVE 'Next Record Retrieved.' TO MFMSGO
  ELSE
    MOVE 'End of File Reached.' TO MFMSGO
  END-IF
```

```
EXEC CICS ENDBR FILE('SKILLFILE') END-EXEC
ELSE
  MOVE 'Error Navigating Forward.' TO MFMSGO
END-IF.
SET SEND-DATAONLY TO TRUE.
PERFORM 900-SEND-PARA.
```

```
*=====*
* SUPPORTING ROUTINES & I/O MAPPING *
*=====*
```

110-RECEIVE-PARA.

```
EXEC CICS RECEIVE
  MAP('MFRNMAP')
  MAPSET('MFRNSET')
  INTO(MFRNMAPI)
  RESP(RESPCODE)
END-EXEC.
```

120-VALIDATE-YEAR.

```
SET VALID-DATA TO TRUE.
IF MFYEARI = SPACES OR LOW-VALUES
  MOVE 'Year is Required.' TO MFMSGO
  SET INVALID-DATA TO TRUE
ELSE IF MFYEARI NOT NUMERIC
  MOVE 'Year Must Be Numeric.' TO MFMSGO
  SET INVALID-DATA TO TRUE
END-IF.
```

150-VALIDATE-INPUT.

```
IF MFCATI NOT = 'S' AND MFCATI NOT = 'P'
  MOVE 'Category must be S (Skills) or P (Projects).'
  TO MFMSGO
  SET INVALID-DATA TO TRUE
END-IF.
```

170-MOVE-MAP-REC-PARA.

```
MOVE MFYEARI TO WS-REC-YEAR.
MOVE MFCATI TO WS-REC-CATEGORY.
MOVE MFTITLEI TO WS-REC-ROLE-TITLE.
MOVE MFSKILLI TO WS-REC-SKILLS.
MOVE MFHL1I TO WS-REC-HL1.
MOVE MFHL2I TO WS-REC-HL2.
MOVE MFHL3I TO WS-REC-HL3.
MOVE MFLINKI TO WS-REC-LINK-REF.
```


180-MOVE-REC-MAP-PARA.

MOVE WS-REC-YEAR TO MFYEARO.
MOVE WS-REC-CATEGORY TO MFCATO.
MOVE WS-REC-ROLE-TITLE TO MFTITLEO.
MOVE WS-REC-SKILLS TO MFSKILLO.
MOVE WS-REC-HL1 TO MFHL1O.
MOVE WS-REC-HL2 TO MFHL2O.
MOVE WS-REC-HL3 TO MFHL3O.
MOVE WS-REC-LINK-REF TO MFLINKO.

900-SEND-PARA.

PERFORM 050-MOVEPARA.
IF SEND-ERASE
EXEC CICS SEND
MAP('MFRNMAP')
MAPSET('MFRNSET')
FROM(MFRNMAPO)
ERASE
END-EXEC
ELSE
EXEC CICS SEND
MAP('MFRNMAP')
MAPSET('MFRNSET')
FROM(MFRNMAPO)
DATAONLY
END-EXEC
END-IF.
SET SEND-DATAONLY TO TRUE.

Tab 5

```

MFRNSET DFHMSD TYPE=MAP,
    MODE=INOUT,
    EXTATT=YES,
    CTRL=(FREEKB,FRSET),
    LANG=COBOL,
    TIOAPFX=YES,
    STORAGE=AUTO
* --- Screen Header ---
MFRNMAP DFHMDI SIZE=(24,80)
    DFHMDF POS=(1,26),
    LENGTH=28,
    ATTRB=(BRT,PROT),
    INITIAL='SKILLS && DETAILS INQUIRY'
    DFHMDF POS=(3,25),
    LENGTH=30,
    ATTRB=(BRT,PROT),
    INITIAL='Marlainna you still Got it!'
* --- Date & Time ---
    DFHMDF POS=(1,58),
    LENGTH=5,
    ATTRB=(NORM,PROT),
    INITIAL='DATE:'
MFDATA DFHMDF POS=(1,64),
    LENGTH=10,
    ATTRB=(BRT,PROT)
    DFHMDF POS=(2,58),
    LENGTH=5,
    ATTRB=(NORM,PROT),
    INITIAL='TIME:'
MFTIME DFHMDF POS=(2,64),
    LENGTH=8,
    ATTRB=(BRT,PROT)

```

* --- Prompts & Input Fields ---

DFHMDF POS=(5,15),	*
LENGTH=12,	*
ATTRB=(NORM,PROT),	*
INITIAL='ENTER YEAR :'	
MFYEAR DFHMDF POS=(5,28),	*
LENGTH=4,	*
ATTRB=(UNPROT,NUM,IC)	
DFHMDF POS=(5,33),	*
LENGTH=1,	*
ATTRB=(ASKIP)	
DFHMDF POS=(5,37),	*
LENGTH=11,	*
ATTRB=(NORM,PROT),	*
INITIAL='CATEGORY :'	
MFCAT DFHMDF POS=(5,49),	*
LENGTH=1,	*
ATTRB=(UNPROT,NORM)	
DFHMDF POS=(5,51),	*
LENGTH=1,	*
ATTRB=(ASKIP)	
DFHMDF POS=(5,53),	*
LENGTH=20,	*
ATTRB=(NORM,PROT),	*
INITIAL='(S=Skills, P=Projects)'	

* --- Divider Line ---

DFHMDF POS=(6,15),	*
LENGTH=55,	*
ATTRB=(NORM,PROT),	*
INITIAL='-----'	

* --- Output Detail Fields ---

DFHMDF POS=(7,15),	*
--------------------	---

LENGTH=12,	*	
ATTRB=(NORM,PROT),		*
INITIAL='ROLE/TITLE :'		
MFTITLE DFHMDF POS=(7,28),		*
LENGTH=40,	*	
ATTRB=(BRT,PROT)		
DFHMDF POS=(7,69),		*
LENGTH=1,	*	
ATTRB=(ASKIP)		
DFHMDF POS=(9,15),		*
LENGTH=12,	*	
ATTRB=(NORM,PROT),		*
INITIAL='SKILLS :'		
MFSKILL DFHMDF POS=(9,28),		*
LENGTH=40,	*	
ATTRB=(BRT,PROT)		
DFHMDF POS=(9,69),		*
LENGTH=1,	*	
ATTRB=(ASKIP)		
DFHMDF POS=(11,15),		*
LENGTH=12,	*	
ATTRB=(NORM,PROT),		*
INITIAL='HIGHLIGHTS :'		
MFHL1 DFHMDF POS=(11,28),		*
LENGTH=40,	*	
ATTRB=(BRT,PROT)		
DFHMDF POS=(11,69),		*
LENGTH=1,	*	
ATTRB=(ASKIP)		
MFHL2 DFHMDF POS=(12,28),		*
LENGTH=40,	*	
ATTRB=(BRT,PROT)		

```

DFHMD F POS=(12,69),      *
LENGTH=1,                 *
ATTRB=(ASKIP)
MFHL3  DFHMD F POS=(13,28),      *
LENGTH=40,                *
ATTRB=(BRT,PROT)
DFHMD F POS=(13,69),      *
LENGTH=1,                 *
ATTRB=(ASKIP)
DFHMD F POS=(15,15),      *
LENGTH=12,                *
ATTRB=(NORM,PROT),        *
INITIAL='LINK/REF  : '
MFLINK DFHMD F POS=(15,28),      *
LENGTH=40,                *
ATTRB=(BRT,PROT)
DFHMD F POS=(15,69),      *
LENGTH=1,                 *
ATTRB=(ASKIP)
* --- Status Message ---
MFMSG  DFHMD F POS=(21,5),      *
LENGTH=70,                *
ATTRB=(BRT,PROT)
* --- Function Key Legend ---
DFHMD F POS=(22,1),        *
LENGTH=78,                *
ATTRB=(NORM,PROT),        *
INITIAL='F1=READ F2=WRITE F3=EXIT F4=UPDATE F5=DELETE'
DFHMD F POS=(23,1),        *
LENGTH=78,                *
ATTRB=(NORM,PROT),        *
INITIAL='F6=CLEAR F7=PREV F8=NEXT F10=CONFIRM UPDATE'

```

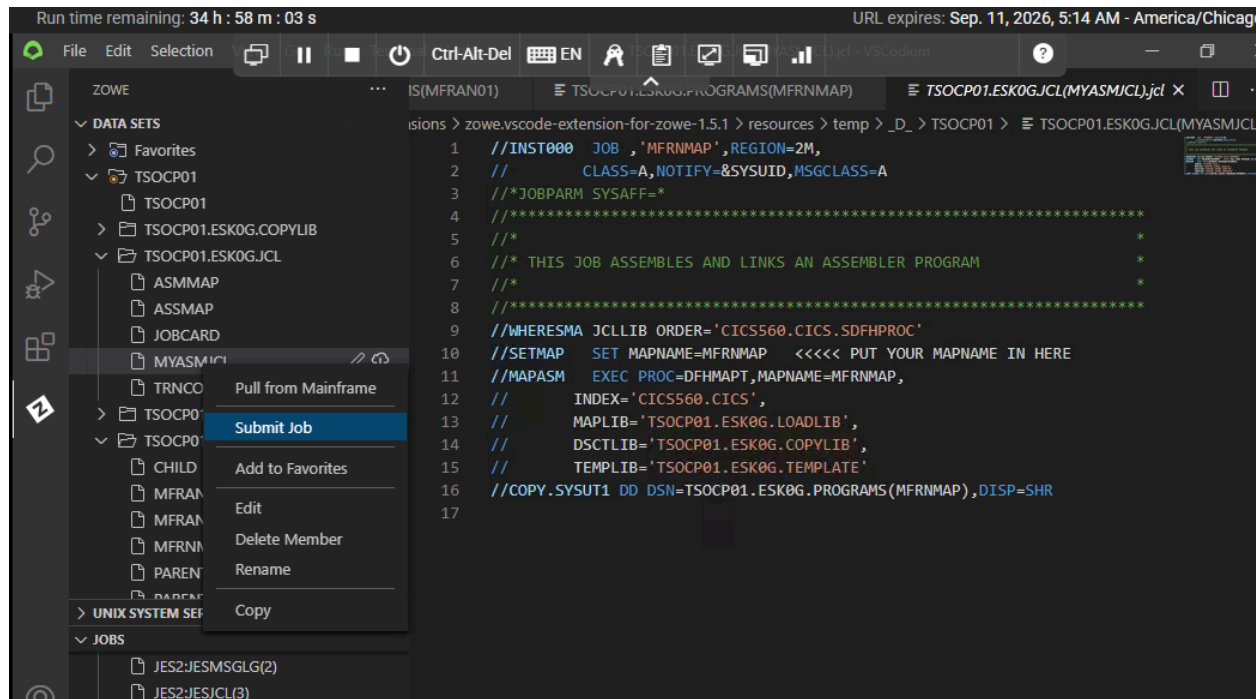
```
DFHMD F POS=(24,1),  
  LENGTH=79,  
  ATTRB=(NORM,PROT),  
  INITIAL='ENTER=BROWSE'  
DFHMSD TYPE=FINAL  
END
```

```
*  
*  
*
```

sTEP TO RUN MAP

NOTE TO self. MAKE EDIT IN VS. MAINFRAME IS A HEADACHE

1. Created my map on text editor
2. Emailed it to my virtual machine
3. Copied the code into visual studio
4. Ran JCL to



Run time remaining: 34 h : 58 m : 03 s URL expires: Sep. 11, 2026, 5:14 AM - America/Chicago

File Edit Selection Ctrl-Alt-Del EN

TSOCPO1.ESK0G.PROGRAMS(MFRNMAP) TSOCPO1.ESK0G.JCL(MYASMJCL).jcl

DATA SETS

- TSOCPO1
- TSOCPO1.ESK0G.COPYLIB
- TSOCPO1.ESK0G.JCL
 - ASMMAP
 - ASSMAP
 - JOBCARD
 - MYASMJCL (selected)
 - TRNCO

Context Menu:

- Pull from Mainframe
- Submit Job (highlighted)
- Add to Favorites
- Edit
- Delete Member
- Rename
- Copy

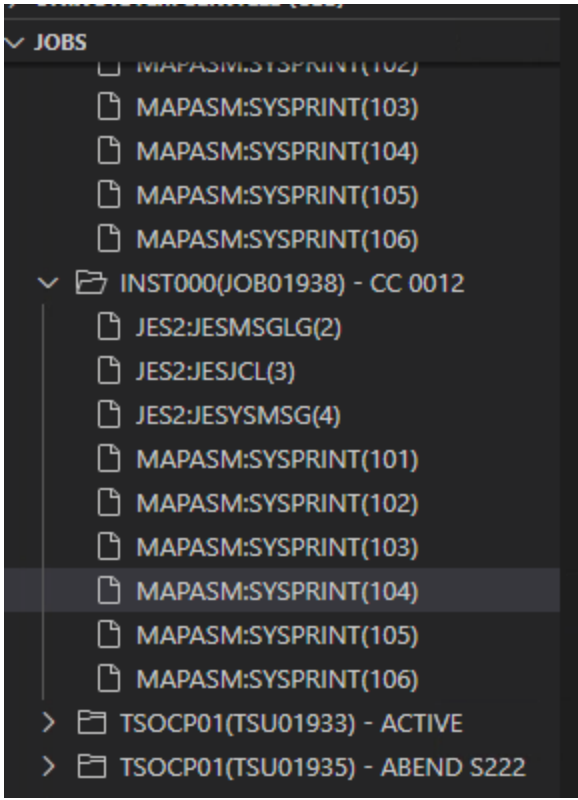
JCL Code:

```
1 //INST000 JOB , 'MFRNMAP', REGION=2M,
2 //      CLASS=A, NOTIFY=&SYSUID, MSGCLASS=A
3 //*JOBPARM SYSAFF=*
4 //*****
5 //*
6 //* THIS JOB ASSEMBLES AND LINKS AN ASSEMBLER PROGRAM
7 //*
8 //*****
9 //WHERE SMA JCLLIB ORDER='CICS560.CICS.SDFHPROC'
10 //SETMAP SET MAPNAME=MFRNMAP <<<<< PUT YOUR MAPNAME IN HERE
11 //MAPASM EXEC PROC=DFHMAPT, MAPNAME=MFRNMAP,
12 //      INDEX='CICS560.CICS',
13 //      MAPLIB='TSOCPO1.ESK0G.LOADLIB',
14 //      DSCTLIB='TSOCPO1.ESK0G.COPYLIB',
15 //      TEMPLIB='TSOCPO1.ESK0G.TEMPLATE'
16 //COPY.SYSUT1 DD DSN=TSOCPO1.ESK0G.PROGRAMS(MFRNMAP), DISP=SHR
17
```

Job1938

ERROR 0012

```
IEF142I INST000 ASMDSECT MAPASM - STEP WAS EXECUTED - COND CODE 0012
IEF285I  TS0CP01.INST000.JOB01938.D0000104.?          SYSOUT
IEF285I  CICS560.CICS.SDFHMAC                        KEPT
IEF285I  VOL SER NOS= ZOS0CA.
IEF285I  SYS1.MACLIB                                  KEPT
IEF285I  VOL SER NOS= ZOS0X0.
IGD105I SYS26248.T020358.RA000.INST000.R0100745      DELETED, DDNAME=SYSUT1
IGD105I SYS26248.T020358.RA000.INST000.R0100746      DELETED, DDNAME=SYSUT2
IGD105I SYS26248.T020358.RA000.INST000.R0100747      DELETED, DDNAME=SYSUT3
IGD104I TS0CP01.ESK0G.COPYLIB                         RETAINED, DDNAME=SYSPUNCH
IGD106I SYS26248.T020358.RA000.INST000.TEMPM.H01     PASSED, DDNAME=SYSIN
IEF373I STEP/ASMDSECT/START 2026248.0203
```

Errors:

** ASMA254I *** MNOTE ***	481+	0,PERFORMANCE OF INPUT MAPPING IS BETTER	01-DFHMD
** ASMA254I *** MNOTE ***	482+	0,WHEN THE FIELDS ARE IN ASCENDING SEQUENCE	01-DFHMD
484 MFDATA DFHMDF POS=(1,64),		*	
LENGTH=10,		*	

```

ATTRB=(BRT,PROT)
** ASMA254| *** MNOTE ***      578+  0,PERFORMANCE OF INPUT MAPPING IS BETTER      01-DFHMD
** ASMA254| *** MNOTE ***      579+  0,WHEN THE FIELDS ARE IN ASCENDING SEQUENCE  01-DFHMD
      581+  PUNCH '      02 MFDATEL  COMP PIC S9(4).' @BBD5E01 01-DFHMD
      582+  PUNCH '      02 MFDATEF  PICTURE X.'    @BBD5E01 01-DFHMD
      583+  PUNCH '      02 FILLER REDEFINES MFDATEF.' @BM13780 01-DFHMD
      584+  PUNCH '      03 MFDATEA  PICTURE X.'    @BM13780 01-DFHMD
      585+  PUNCH '      02 FILLER  PICTURE X(4).'    @L3C 01-DFHMD
      586+  PUNCH '      02 MFDATEI  PIC X(10).'     @BBD5E01 01-DFHMD
      587  DFHMDF POS=(2,58),
           LENGTH=5,
           ATTRB=(NORM,PROT),
           INITIAL='TIME:'
** ASMA254| *** MNOTE ***      681+  0,PERFORMANCE OF INPUT MAPPING IS BETTER      01-DFHMD
** ASMA254| *** MNOTE ***      682+  0,WHEN THE FIELDS ARE IN ASCENDING SEQUENCE  01-DFHMD
      684 MFTIME  DFHMDF POS=(2,64),
           LENGTH=8,
           ATTRB=(BRT,PROT)
** ASMA254| *** MNOTE ***      778+  0,PERFORMANCE OF INPUT MAPPING IS BETTER      01-DFHMD
** ASMA254| *** MNOTE ***      779+  0,WHEN THE FIELDS ARE IN ASCENDING SEQUENCE  01-DFHMD
      781+  PUNCH '      02 MFTIMEL  COMP PIC S9(4).' @BBD5E01 01-DFHMD
      782+  PUNCH '      02 MFTIMEF  PICTURE X.'    @BBD5E01 01-DFHMD
      783+  PUNCH '      02 FILLER REDEFINES MFTIMEF.' @BM13780 01-DFHMD
      784+  PUNCH '      03 MFTIMEA  PICTURE X.'    @BM13780 01-DFHMD
      785+  PUNCH '      02 FILLER  PICTURE X(4).'    @L3C 01-DFHMD
      786+  PUNCH '      02 MFTIMEI  PIC X(8).'     @BBD5E01 01-DFHMD
i      787 * --- Prompts & Input Fields ---

```

Page 3

Active Usings: None

Loc	Object Code	Addr1	Addr2	Stmt	Source Statement
788					DFHMDF POS=(5,15),

HLASM R6.0 2026/09/05 02.03

NEXT GROUP OF ERRORS

```

00C          0000C          206+MFRNMAP DS      0C .      MAP ORIG
                                207+DFHL1 EQU      *      START OF
                                208+      DC      AL2(DFHK1)      MAP DESC
                                209+      DC      XL2('0000')      INPUT PA
                                210+      DC      XL2('0000')      OUTPUT P
                                211+      DC      XL2('0000')      ACTIVE P
                                212+      DC      CL8('MFRNMAP' .      MAP
                                213+      DC      AL2(DFHM1)      MAP LENG
ASMA044E Undefined symbol - DFHM1
ASMA435I Record 1171 in CICS560.CICS.SDFHMAC(DFHMDI) on volume: Z0S0CA
01E 0000      214+      DC      AL2(DFHS1)      MAP SPEC
ASMA044E Undefined symbol - DFHS1
ASMA435I Record 1172 in CICS560.CICS.SDFHMAC(DFHMDI) on volume: Z0S0CA
020 0000      215+      DC      AL2(DFHN1)      INPUT WO
ASMA044E Undefined symbol - DFHN1
ASMA435I Record 1173 in CICS560.CICS.SDFHMAC(DFHMDI) on volume: Z0S0CA
022 0000      216+      DC      AL2(DFHP1)      OUTPUT W
ASMA044E Undefined symbol - DFHP1
ASMA435I Record 1174 in CICS560.CICS.SDFHMAC(DFHMDI) on volume: Z0S0CA
024 C0      217+      DC      AL1(192)      MAP
025 C3      218+      DC      CL1('C'      3270 WRI
026 0000      219+      DC      AL2(DFHQ1)      3270 CUR
ASMA044E Undefined symbol - DFHQ1
ASMA435I Record 1177 in CICS560.CICS.SDFHMAC(DFHMDI) on volume: Z0S0CA
028 00      220+      DC      AL1(0)      MAP MARG
029 18      221+      DC      AL1(24)      MAP LENG
02A 50      222+      DC      AL1(80)      MAP WIDT
F1=HELP      F2=SPLIT      F3=END      F4=RETURN      F5=IFIND      F6=BOOK
F7=UP      F8=DOWN      F9=SWAP      F10=LEFT      F11=RIGHT      F12=RETRIEVE

```

COPIED RESULTS INTO AI

Drafted FIX

pASTED FIX CODE FOR maP INTO **TSOCP01.ESK0G.PROGRAMS(MFRNMAP)**

jOB 1942

Fix: Created the missing fields

JOB 1944 RC = 8

Issue: the continuation character must be in column 72

Fix move

```
> Users > IBM > .vscode-oss > extensions > zowe.vscode-extension-for-zowe-1.5.1 > resources > temp > _D_ > TSOCPO1 > TSOCPO1.ESKOG.PROGRAMS(MFRNMAP)
1  MFRNSET  DFHMSD TYPE=MAP,
2          MODE=INOUT,
3          EXTATT=YES,
4          CTRL=(FREEKB,FRSET),
5          LANG=COBOL,
6          TIOAPFX=YES,
7          STORAGE=AUTO
8  * --- Screen Header ---
9  MFRNMAP  DFHMDI SIZE=(24,80),Line=1
10          DFHMDF POS=(1,26),
11          LENGTH=28,
12          ATTRB=(BRT,PROT),
13          INITIAL='SKILLS && DETAILS INQUIRY'
14          DFHMDF POS=(3,25),
15          LENGTH=30,
16          ATTRB=(BRT,PROT),
17          INITIAL='Marlainna you still Got it!'
18  * --- Date & Time ---
19          DFHMDF POS=(1,58),
20          LENGTH=5,
21          ATTRB=(NORM,PROT),
22          INITIAL='DATE: '
23  MFDATE   DFHMDF POS=(1,64),
24          LENGTH=10,
25          ATTRB=(BRT,PROT)
26          DFHMDF POS=(2,58),
27          LENGTH=5,
28          ATTRB=(NORM,PROT),
29          INITIAL='TIME: '

```

Ln 1, Col 73 (1 selected) Spaces: 6 UTF-8 CRLF Plain Text

Drafts (12... (no subie... TSOCPO1... Session A... Session B... Session C... FSP 10:53 PM

Job1945

accomplished

09/05 - That was crazy

Key to Note: Transferring BMS maps between local text files and z/OS datasets often introduces line-endings or mainframe EBCDIC translation issues.

I got a solid amount of foundation work done tonight! Here is a summary of what was accomplished and the exact checklist waiting for you when you pick this back up:

What You Completed Tonight

- **BMS Map Source:** Aligned all field names in your BMS map and program copybook.
- **Map Assembly:** Ran the JCL to assemble MFRNSET INTO MFRNMAP
- **VSAM KSDS Creation:** Created your VSAM KSDS dataset TSOCP01.ESK0G.SKILLS with a primary key length of 4 (KEYS(4 0)), record length of 240 bytes, and proper space allocation (CYLINDERS(1 1)).
- **CICS Compilation JCL:** Resolved initial z/OSMF / Zowe submission formatting errors (JCL JOB card slash fixes).

Your To-Do List for Next Session

1.Update Transaction ID in COBOL Source:1 min.

Shorten TRANSID('MYTRN') to a 4-character ID (e.g., TRANSID('MYTR')) in MFRNMAPP.cbl to clear the pre-translator warning (DFH7225I).

2.Get Clean COBOL Compilation (MAXCC=0):2 min.

Verify that TSOCP01.ESK0G.COPYLIB is included under //SYSLIB DD in your COBOL compile step so the compiler finds COPY MFRNSET. and resolves all IGYPS2121-S errors.

1. MAPNAME:MFRNMAP

TSOCP01.ESK0G.PROGRAMS(MFRNMAP)

```

> Users > IBM > .vscode-oss > extensions > zowe.vscode-extension-for-zowe-1.5.1 > resources > temp >
1  MFRNMAP  DFHMSD TYPE=MAP,MODE=INOUT,EXTATT=YES,LANG=COBOL,TIOAPFX=YES
2  * --- Screen Header ---
3  MFRNSET  DFHMDI SIZE=(24,80)
4          DFHMDI POS=(1,26),LENGTH=28,ATTRB=(BRT,PROT),
5          INITIAL='SKILLS && DETAILS INQUIRY'
6  * --- Date & Time ---
7          DFHMDI POS=(1,58),LENGTH=5,ATTRB=(NORM,PROT),INITIAL='DATE:'
8  MFDAT   DFHMDI POS=(1,64),LENGTH=10,ATTRB=(BRT,PROT)

```

```

MFRNSET  DFHMSD TYPE=MAP,MODE=INOUT,EXTATT=YES,LANG=COBOL,TIOAPFX=YES
* --- Screen Header ---
MFRNMAP  DFHMDI SIZE=(24,80)
          DFHMDI POS=(1,26),LENGTH=28,ATTRB=(BRT,PROT),
          INITIAL='SKILLS && DETAILS INQUIRY'
* --- Date & Time ---
          DFHMDI POS=(1,58),LENGTH=5,ATTRB=(NORM,PROT),INITIAL='DATE:'

```

Run `jcl` Ran the JCL to assemble `MFRNSET` into `MFRNMAP`.

```
Users > IBM > .vscode-oss > extensions > zowe.vscode-extension-for-zowe-1.5.1 > resources > temp > _D_
1  //INST000 JOB , 'MFRNMAP', REGION=2M,
2  //          CLASS=A, NOTIFY=&SYSUID, MSGCLASS=A
3  // *JOBPARM SYSAFF=*
4  //*****
5  //*
6  //* THIS JOB ASSEMBLES AND LINKS AN ASSEMBLER PROGRAM
7  //*
8  //*****
9  //WHERE SMA JCLLIB ORDER='CICS560.CICS.SDFHPROC'
10 //SETMAP SET MAPNAME=MFRNMAP <<<<< PUT YOUR MAPNAME IN HERE
11 //MAPASM EXEC PROC=DFHMAPT, MAPNAME=MFRNMAP,
12 //      INDEX='CICS560.CICS',
13 //      MAPLIB='TSOCP01.ESK0G.LOADLIB',
14 //      DSCTLIB='TSOCP01.ESK0G.COPYLIB',
15 //      TEMPLIB='TSOCP01.ESK0G.TEMPLATE'
16 //COPY.SYSUT1 DD DSN=TSOCP01.ESK0G.PROGRAMS(MFRNMAP), DISP=SHR
17
```

JOB1974 CC-000

2. CHECK FOR THE COPY BOOK

FOUND CopyBOOK NAME : MFRNMAP

TSOCP01.ESK0G.COPYLIB (MFRNMAP)

```
File Edit Selection View Go Run Terminal Help
ZOWE
DATA SETS
  Favorites
  TSOCPO1
    TSOCPO1
    TSOCPO1.ESK0G.COPYLIB
      MFRAN01
      MFRNMAP
      STUDMAP
    TSOCPO1.ESK0G.JCL
      ASMMAP
      ASSMAP
      JOBCARD
      MYASMJCL
      MYCICJCL
      TRNCOBLK
      VSAMJCL
TSOCPO1.ESK0G.JCL(MYCICJCL).jcl
c:\Users\IBM>.vscode-oss>extensions>zowe.vscode-extension
54
55
56
57
58
59
60
61
62
63
64
65
66
67
68
69
70
03 MFHL3A PICTURE X.
02 FILLER PICTURE X(4).
02 MFHL3I PIC X(40).
02 MFLINKL COMP PIC S9(4).
02 MFLINKF PICTURE X.
02 FILLER REDEFINES MFLINKF.
03 MFLINKA PICTURE X.
02 FILLER PICTURE X(4).
02 MFLINKI PIC X(40).
02 MFMSGI COMP PIC S9(4).
02 MFMSGF PICTURE X.
02 FILLER REDEFINES MFMSGF.
03 MFMSGA PICTURE X.
02 FILLER PICTURE X(4).
02 MFMSGI PIC X(70).
01 MFRNMAP0 REDEFINES MFRNMAP1.
02 FILLER PIC X(12).
```

**3. COMPILE PROGRAM USING
COPYBOOK
TSOCPO1.ESK0G.JCL(MYCICJCL)**

POINT TO MY COPYBOOK

TSOCPO1.ESK0G.COPYLIB (MFRNMAP)

```
Welcome to Z Open Editor  TSOC01.ESK0G.PROGRAMS(MFRNMAP)  TSOC01.ESK0G.PROGRAMS(MFRNMAP)
c: > Users > IBM > .vscode-oss > extensions > zowe.vscode-extension-for-zowe-1.5.1 > resources > temp > _D_ > TSOC01
1  //INST000 JOB , 'MFRNMAP', REGION=2M,
2  //      CLASS=A, NOTIFY=&SYSUID, MSGCLASS=A
3  /**JOBPARM SYSAFF=*
4  /*******
5  /***
6  /*** THIS JOB ASSEMBLES AND LINKS AN ASSEMBLER PROGRAM
7  /***
8  /*******
9  //WHERESMA JCLLIB ORDER='CICS560.CICS.SDFHPROC'
10 //SETMAP  SET MAPNAME=MFRNMAP  <<<<< PUT YOUR MAPNAME IN HERE
11 //MAPASM  EXEC PROC=DFHMAPT, MAPNAME=MFRNMAP,
12 //      INDEX='CICS560.CICS',
13 //      MAPLIB='TSOC01.ESK0G.LOADLIB',
14 //      DSCTLIB='TSOC01.ESK0G.COPYLIB',
15 //      TEMPLIB='TSOC01.ESK0G.TEMPLATE'
16 //COPY.SYSUT1 DD DSN=TSOC01.ESK0G.PROGRAMS(MFRNMAP), DISP=SHR
17
```

**4. CONFIRM VSAM File nick name
in program:**

```
MOVE MFYEAR1 TO WS-REC-YEAR.  
  
EXEC CICS  
  READ FILE('SKILLS')  
  INTO(WS-SKILL-DETAIL-RECORD)  
  RIDFLD(WS-REC-YEAR)  
  RESP(RESPCODE)  
END-EXEC.  
  
IF RESPCODE = DFHRESP(NORMAL)
```

errorS:

Missing copybook

Errors

904 IGYPS2121-S "MFCAT1" was not defined as a data-name. The statement was discarded.

Same message on line: 904 912

913 IGYPS2121-S "MFTITLE1" was not defined as a data-name. The statement was discarded.

914 IGYPS2121-S "MFSKILL1" was not defined as a data-name. The statement was discarded.

915 IGYPS2121-S "MFHL11" was not defined as a data-name. The statement was discarded.

916 IGYPS2121-S "MFHL2I" was not defined as a data-name. The statement was discarded.

917 IGYPS2121-S "MFHL3I" was not defined as a data-name. The statement was discarded.

918 IGYPS2121-S "MFLINKI" was not defined as a data-name. The statement was discarded.

921 IGYPS2121-S "MFYEARO" was not defined as a data-name. The statement was discarded.

922 IGYPS2121-S "MFCATO" was not defined as a data-name. The statement was discarded.

923 IGYPS2121-S "MFTITLEO" was not defined as a data-name. The statement was discarded.

924 IGYPS2121-S "MFSKILLO" was not defined as a data-name. The statement was discarded.

1PP 5655-EC6 IBM Enterprise COBOL for z/OS 6.3.0 P191120 MFRNMAPP Date
09/05/2026 Time 06:15:36 Page 40

0 925 IGYPS2121-S "MFHL1O" was not defined as a data-name. The statement was discarded.

926 IGYPS2121-S "MFHL2O" was not defined as a data-name. The statement was discarded.

927 IGYPS2121-S "MFHL3O" was not defined as a data-name. The statement was discarded.

928 IGYPS2121-S "MFLINKO" was not defined as a data-name. The statement was discarded.

-Messages	Total	Informational	Warning	Error	Severe	Terminating
0Printed:	84	3		81		

Checked copybook in program

Ran the JCL to assemble MFRNMAP into MFRNSET.

Map set : MFRNMAP

Map name: MFRNSET in my program is it WORKING-STORAGE SECTION.

WORKING-STORAGE SECTION structure is configured correctly!

DATA DIVISION.

WORKING-STORAGE SECTION.

*--- BMS MAP COPYBOOK ---

COPY MFRNSET.

*--- DFHAID & DFHBMSCA COPYBOOKS ---

COPY DFHAID.

COPY DFHBMSCA.

*--- VSAM RECORD STRUCTURE (MATCHES RECORD LENGTH)

01 WS-SKILL-DETAIL-RECORD.

05 WS-REC-YEAR PIC X(04).

05 WS-REC-CATEGORY PIC X(01).

05 WS-REC-ROLE-TITLE PIC X(40).

05 WS-REC-SKILLS PIC X(40).

05 WS-REC-HIGHLIGHTS.

10 WS-REC-HL1 PIC X(40).

10 WS-REC-HL2 PIC X(40).

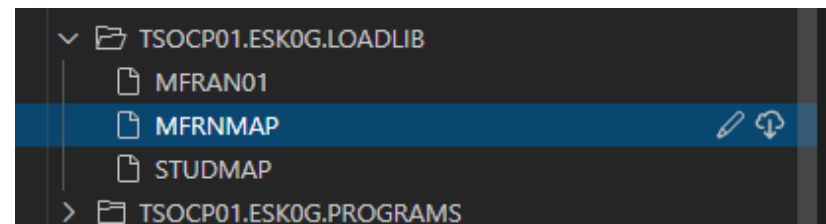
10 WS-REC-HL3 PIC X(40).

05 WS-REC-LINK-REF PIC X(40).

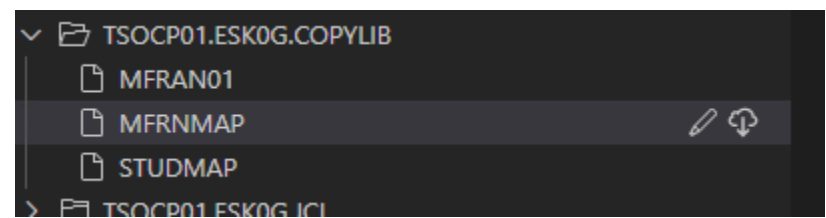
05 FILLER PIC X(35) VALUE

So the copy book I am suppose to be using is

LOAD DATASET



COPYBOOK DATASET



Fix: updated mapname

```
DATA DIVISION.  
  
WORKING-STORAGE SECTION.  
*--- BMS MAP COPYBOOK ---  
COPY MFRNMAP [  
  
*--- DFHAID & DFHBMSCA COPYBOOKS ---  
COPY DFHAID.
```

Update transaction ID: MFTR

```

94         PERFORM 900-SEND-PARA
95         END-EVALUATE
96     END-IF.
97
98     EXEC CICS RETURN
99         TRANSID('MFTR')
100         COMMAREA(WS-COMM-AREA)
101         LENGTH(LENGTH OF WS-COMM-AREA)
102     END-EXEC.
103
104

```

How to Fix

Change line 13 in your JCL from actual program member **MFRNMAPP**:

```
//SYSIN DD DISP=SHR,DSN=TSOCP01.ESK0G.PROGRAMS(MFRNMAPP)
```

Verification Steps

1. Update line 13 to point to **TSOCP01.ESK0G.PROGRAMS(MFRNMAPP)**.
2. Double-check line 21 (under **//SYSLIB DD**) to ensure **TSOCP01.ESK0G.COPYLIB** is listed so **COPY MFRNSET.** can resolve.
3. Save (**Ctrl + S**) and resubmit the job.

4.

COMPILE MY CICS PROGRAM

1. Update the CPYBOOK and LINK library
2. Run my JCL: TSOCP01.ESK0g.JCL(MYCICJCL)

```
//TSOCP01X JOB 30000000,'ESK0 JOB CARD ',MSGLEVEL=(1,1),          00010000
//  CLASS=A,MSGCLASS=Q,NOTIFY=&SYSUID,REGION=0M                  00020000
//***** 00030000
//*                                * 00040000
//***** 00050000
//TRN   EXEC PGM=DFHECP1$,REGION=0M,
//      PARM=(CICS)
//STEPLIB DD DSN=CICS560.CICS.SDFHLOAD,DISP=SHR
//SYSPRINT DD SYSOUT=*
//SYSPUNCH DD DSN=&&SYSCIN,DISP=(,PASS),
//      SPACE=(CYL,(1,1)),
//      DCB=(RECFM=FB,LRECL=80,BLKSIZE=400)
//SYSIN   DD DISP=SHR,DSN=TSOCP01.ESK0G.PROGRAMS(MFRNMAPP)
//*----- EXEC the COBOL II Compiler. -----
//COB    EXEC PGM=IGYCRCTL,REGION=0M,
//      COND=(7,LT),
//      PARM=(APOST,NODYNAM,OPTIMIZE,SSRANGE,
//      LIST,MAP,XREF,'NOCOMPILE(E)',LIB,RENT)
//STEPLIB DD DSN=IGY.SIGYCOMP,DISP=SHR
//SYSPRINT DD SYSOUT=*
//SYSLIB  DD DSN=CICS560.CICS.SDFHCOB,DISP=SHR
```

```
//      DD DSN=CICS560.CICS.SDFHSAMP,DISP=SHR
//      DD DSN=TSOCP01.ESK0G.COPYLIB,DISP=SHR
//SYSUT1 DD SPACE=(CYL,(1,1))
//SYSUT2 DD SPACE=(CYL,(1,1))
//SYSUT3 DD SPACE=(CYL,(1,1))
//SYSUT4 DD SPACE=(CYL,(1,1))
//SYSUT5 DD SPACE=(CYL,(1,1))
//SYSUT6 DD SPACE=(CYL,(1,1))
//SYSUT7 DD SPACE=(CYL,(1,1))
//SYSUT8 DD SPACE=(CYL,(1,1))
//SYSUT9 DD SPACE=(CYL,(1,1))
//SYSUT10 DD SPACE=(CYL,(1,1))
//SYSUT11 DD SPACE=(CYL,(1,1))
//SYSUT12 DD SPACE=(CYL,(1,1))
//SYSUT13 DD SPACE=(CYL,(1,1))
//SYSUT14 DD SPACE=(CYL,(1,1))
//SYSUT15 DD SPACE=(CYL,(1,1))
//SYSMDECK DD SPACE=(CYL,(1,1))
//SYSIN DD DSN=&&SYSCIN,DISP=(OLD,PASS)
//SYSLIN DD DSN=&&OBJMOD,DISP=(,PASS),
//      SPACE=(CYL,(1,1)),
//      DCB=(RECFM=FB,LRECL=80,BLKSIZE=400)
//*----- LinKEDit the output into the specified Load Library -----
//LKED EXEC PGM=IEWL,REGION=2048K,
//      COND=(7,LT),
//      PARM=(LIST,XREF)
//SYSPRINT DD SYSOUT=*
//SYSLIB DD DSN=CICS560.CICS.SDFHLOAD,DISP=SHR
```

```
//      DD DSN=CEE.SCEELKED,DISP=SHR
//SYSUT1 DD SPACE=(CYL,(1,1))
//SYSLIN DD DSN=CICS560.CICS.SDFHCOB(DFHEILIC),DISP=SHR,
//      DCB=BLKSIZE=400
//      DD DSN=&&OBJMOD,DISP=(OLD,PASS)
//      DD DDNAME=SYSIN
//SYSLMOD DD DISP=SHR,DSN=TSOCP01.ESK0G.LOADLIB
//SYSIN DD *
      NAME PAYPGMM(R)
/*
/*----- Delete any temporary datasets -----
//DELTEMP EXEC PGM=IEFBR14
//DD1      DD DSN=&&SYSCIN,DISP=(OLD,DELETE)
//DD2      DD DSN=&&OBJMOD,DISP=(OLD,DELETE)
```

JOB 1974 ran successfully CC0000

5.

Step 1: Check Where MFRNSET Was Saved

Open Zowe Explorer or ISPF Option 3.4 and expand your PDS datasets:

1. Look inside **TSOCP01.ESK0G.COPYLIB** (or whichever copybook PDS you created).
2. Is member **MFRNSET** listed inside it?
 - If NO: Your BMS assembly job saved the copybook to a different dataset or member name. Check the **//SYSPUNCH DD** statement in your BMS assembly JCL to see where **MFRNSET** was written.
 - If YES: Move to Step 2.

Step 2: Update the Compile JCL SYSLIB Concatenation

In your compilation JCL (TRNCOBLK), look at the **//COB EXEC PGM=IGYCRCTL** step and inspect the **//SYSLIB DD** statements. Ensure the

dataset containing MFRNSET is explicitly listed:

My COPYBOOK SHOULD BE MY MAPNAME THAT COMPILED SUCCESSFULLY, HAVE TO UPDATE MY COPYBOOK NAME

Can check my resources

3.Initialize VSAM Dataset (Write Dummy Record):1 min.

Run a quick IDCAMS REPRO or COBOL batch load to put an initial record into TSOCP01.ESK0G.SKILLS so CICS can open the file without empty-file errors.

4.Define Resources in CICS (CEDA):3 min.

Run CEDA commands to define and install your resources in CICS:

- CEDA DEFINE PROGRAM(MFRNMAPP) GROUP(MYGRP)
- CEDA DEFINE MAPSET(MFRNSET) GROUP(MYGRP)

- CEDA DEFINE TRANSACTION(MYTR) PROGRAM(MFRNMAPP) GROUP(MYGRP)
- CEDA DEFINE FILE(SKILLFILE) DSNAME(TSOCP01.ESK0G.SKILLS) GROUP(MYGRP)
- CEDA INSTALL GROUP(MYGRP)

5.Test Application in CICS:Testing.

Clear your CICS screen, type MYTR, and test your BMS map display and VSAM CRUD functions.

Find my region

To find your active CICS region name (also known as the **APPLID** or **SYSID**), you can check a few direct places in CICS or ISPF:

1. Check the Bottom Right of Your CEDA Screen

Look at the CEDA screen you were just on (from your earlier screenshot). In the bottom right corner, CICS explicitly displays the active region details:

- **APPLID=ES1ACICS** $\rightarrow$ This is the actual name of your active CICS region.
- **SYSID=ESK0** $\rightarrow$ This is the local CICS system ID.

CEDA DI G(*)

```

DI G(*)
ENTER COMMANDS
GROUP
ADNPCRGP      —
AUTOGRP
CEE
CICSA
CICSU
CMA$EXBS
CONSOLSN
DFH$ACCT
DFH$AFLA
DFH$AXIS
DFH$BABR
DFH$BARF
DFH$BAT
DFH$BMSP
DFH$BR
DFH$BRLK
+ DFH$CBTS

                                                                    SYSID=ESK0 APPLID=ES1ACICS
RESULTS: 1 TO 17                      DSN=ITCICSA.CICS560.DFHCSO
PF 1 HELP      3 END 4 TOP 5 BOT 6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF 12 CNCL

```

Check the Bottom Right of Your CEDA Screen

Look at the CEDA screen you were just on (from your earlier screenshot). In the bottom right corner, CICS explicitly displays the active region details:

- **APPLID=ES1ACICS** This is the actual name of your active CICS region.
- **SYSID=ESK0** This is the local CICS system ID.

2. Run the CEMT Inquire Command

CEDA inquire system

Clear your CICS screen (press **Clear** or **Ctrl + R** depending on your emulator setup), type the following command, and press **Enter**:

```

INQUIRE SYSTEM
STATUS: RESULTS - OVERTYPE TO MODIFY
  System
  Aging( 00001 )
  Aidcount(00000)
  Akp( 04000 )
  Cicstslevel(050600)
  Cmdprotect(Cmdprot)
  Db2conn(ES1UDB2)
  Debugtool( Nodebug )
  Dfltuser(CICSSTC)
  Dsalimit( 08388608 )
  Dsrtprogram( NONE )
  Dtrprogram( DFHDYP )
  Dumping( Nosysdump )
  Edsalimit( 0251658240 )
  Forceqr( Noforce )
  Lastcoldtime( )
  Lastemertime( )
+ Lastinittime(08/22/26 10:20:20)

                                SYSID=ESK0 APPLID=ES1ACICS
  RESPONSE: NORMAL                TIME: 14.51.33  DATE: 09/05/26
PF 1 HELP      3 END      5 VAR      7 SBH 8 SFH 9 MSG 10 SB 11 SF

```

When you run **CEDA DEFINE** and **CEDA INSTALL**, CICS will automatically install your resources directly into this **ES1ACICS** region.

cONNECT TO cics

Blank terminal screen, define and install your resources into a NEW group **ESK0G**.

```
CEDA DI g(ESK0G) _
```

```

DI G(ESK0G)
OVERTYPE TO MODIFY
CEDA Display
  Group      ==> ESK0G
  LISt       ==>
  ALl        ==> *
  ATomservice ==>
  Bundle     ==>
  CONnection ==>
  CORbaserver ==>
  DB2Conn    ==>
  DB2Entry   ==>
  DB2Tran    ==>
  DJar       ==>
  DOctemplate ==>
  DUmpcode   ==>
  Enqmodel   ==>
  File       ==>
  Ipconn     ==>
+ JOurnalmodel ==>
S Group ESK0G not found.

                                SYSID=ESK0 APPLID=ES1ACICS
DISPLAY UNSUCCESSFUL          DSN=ITCICSA.CICS560.DFHCSO
PF 1 HELP                     3 END
                                6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF 12 CNCL

```

CONFIRMS THE GROUP IS NOT FOUND. I WILL DEFINE IT NOW

First, left view all the practice resources in our group using CEDA DISPLAY:

CEDA DI G(ESK0)

Vista TN3270 Session C

File Edit Font Transfer Macro Options Window Help

1 2 3 4 5

DI G(T8)

ENTER	COMMANDS			DATE	TIME
NAME	TYPE	GROUP			
+ LGN1	TRANSACTION	T8		21.347	02.19.17
MENU	TRANSACTION	T8		21.344	03.00.53
MNU1	TRANSACTION	T8		21.347	02.20.00
NEWC	TRANSACTION	T8		21.345	02.40.02
TRN1	TRANSACTION	T8		21.344	03.00.30
TRN2	TRANSACTION	T8		21.344	03.00.30
TRN3	TRANSACTION	T8		21.344	03.00.42
ABC	DB2ENTRY	T8		21.345	02.32.48
LGN2	DB2ENTRY	T8		21.347	02.22.40
LGN3	DB2TRAN	T8		21.347	02.23.19

SYSDI=CICS APPLID=CICS

All the menu set, all the corresponding programs, all the

Lets add a resource: MAPSET, PROGRAM:MFRNMAPP,
TRANID:MFTR

CEDA DEF MAPSET(MFRNSET) G(ESK0G)

```
DEF MAPSET(MFRNSET) G(ESK0G)
OVERTYPE TO MODIFY                                CICS RELEASE = 0730
CEDA DEFINE Mapset( MFRNSET )
  Mapset      : MFRNSET
  Group       : ESK0G
  Description ==>
  Resident    ==> No           No | Yes
  Usage       ==> Normal       Normal | Transient
  UseIcopy    ==> No           No | Yes
  Status      ==> Enabled      Enabled | Disabled
  RSL         : 00            0-24 | Public
DEFINITION SIGNATURE
  DEFINetime   : 09/05/26 15:31:51
  CHANGETime   : 09/05/26 15:31:51
  CHANGEUsrid  : CICSSTC
  CHANGEAGent  : CSDApi       CSDApi | CSDBatch
  CHANGEAGRel  : 0730

I New group ESK0G created.

DEFINE SUCCESSFUL                                SYSID=ESK0 APPLID=ES1ACICS
DSN=ITCICSA.CICS560.DFHCSO
PF 1 HELP 2 COM 3 END                            6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF 12 CNCL
```

PF3

PF3

NOW INSTALL

CEDA INS MAPSET(MFRNSET) G(ESK0G)

```

INS MAPSET(MFRNSET) G(ESK0G)
OVERTYPE TO MODIFY
CEDA Install
  ATomservice ==>
  Bundle ==>
  CONnection ==>
  CORbaserver ==>
  DB2Conn ==>
  DB2Entry ==>
  DB2Tran ==>
  DJar ==>
  DOctemplate ==>
  DUmpcode ==>
  Enqmodel ==>
  File ==>
  Ipconn ==>
  JOurnalmodel ==>
  JVmserver ==>
  LIBrary ==>
+ LSRpool ==>

                                SYSID=ESK0 APPLID=ES1ACICS
INSTALL SUCCESSFUL             DSN=ITCICSA.CICS560.DFHCS
PF 1 HELP                      3 END                                6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF 12 CNCL

```

CEDA DEF PROG(MFRNMAPP) G(ESK0G)

PF3

PF3

```
DEF MAPSET(MFRNMAPP) G(ESK0G)
OVERTYPE TO MODIFY                                CICS RELEASE = 0730
CEDA DEFine Mapset( MFRNMAPP )
  MApset      : MFRNMAPP
  Group       : ESK0G
  DEScription ==> ■
  RESident    ==> No                No | Yes
  USAge       ==> Normal            Normal | Transient
  USElpacopy  ==> No                No | Yes
  Status      ==> Enabled          Enabled | Disabled
  RSL         : 00                0-24 | Public
DEFINITION SIGNATURE
  DEFInetime  : 09/05/26 15:33:27
  CHANGETime  : 09/05/26 15:33:27
  CHANGEUsrid : CICSSTC
  CHANGEAGEnt : CSDApi            CSDApi | CSDBatch
  CHANGEAGRel : 0730

                                           SYSID=ESK0 APPLID=ES1ACICS
DEFINE SUCCESSFUL                        DSN=ITCICSA.CICS560.DFHCS
PF 1 HELP 2 COM 3 END                    6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF 12 CNCL
```

NOW INSTALL

CEDA INS PROG(MFRNMAPP) G(ESK0G)

```
INS MAPSET(MFRNMAPP) G(ESK0G)
OVERTYPE TO MODIFY
CEDA Install
  ATomservice ==> ■
  Bundle ==>
  CONnection ==>
  CORbaserver ==>
  DB2Conn ==>
  DB2Entry ==>
  DB2Tran ==> [ ]
  DJar ==>
  DOctemplate ==>
  DUmpcode ==>
  Enqmodel ==>
  File ==>
  Ipconn ==>
  JOurnalmodel ==> [ ]
  JVmserver ==>
  LIBrary ==>
+ LSRpool ==>

                                SYSID=ESK0 APPLID=ES1ACICS
INSTALL SUCCESSFUL             DSN=ITCICSA.CICS560.DFHCSO
PF 1 HELP      3 END          6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF 12 CNCL
```

**CEDA DEF TRANS(MYTR) PROG(MFRNMAPP)
G(ESK0G)**

PF3

PF3

```

DEF TRANS(MYTR) PROG(MFRNMAPP) G(ESK0G)
OVERTYPE TO MODIFY
CEDA DEFINE TRANSAction( MYTR )
  TRANSAction      : MYTR
  Group            : ESK0G
  DEScription      ==> ■
  PROGram          ==> MFRNMAPP
  TWAsize          ==> 00000          0-32767
  PROFile          ==> DFHCICST
  PArtitionset     ==>
  STATUS           ==> Enabled       Enabled | Disabled
  PRIMedsize       : 00000          0-65520
  TASKDATALoc      ==> Any           Below | Any
  TASKDATAKey      ==> User          User | Cics
  STOrageclear      ==> No           No | Yes
  RUNaway          ==> System        System | 0 | 250-2700000
  SHUTDOWN         ==> Disabled      Disabled | Enabled
  ISolate          ==> Yes           Yes | No
  Brexit           ==>
+ REMOTE ATTRIBUTES

                                                                    SYSID=ESK0 APPLID=ES1ACICS
DEFINE SUCCESSFUL          DSN=ITCICSA.CICS560.DFHCSD
PF 1 HELP 2 COM 3 END      6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF 12 CNCL

```

NOW INSTALL
CEDA INS TRANS(MYTR) G(ESK0G)

041022

NOW LETS INSTALL THE NICKNAME FOR THE KSDS
FILE TSOCP01.ESK0G.SKILLS

CEDA DEF FILE(SKILLS) G(ESK0G)

Type in the actual DATASET NAME

```

DEF FILE(SKILLS) G(ESK0G)
OVERTYPE TO MODIFY
CEDA Define File( SKILLS )
File      : SKILLS
Group     : ESK0G
DEScriptio ==>
VSAM PARAMETERS
DSName    ==> tsocp01.esk0g.skills
Password  : PASSWORD NOT SPECIFIED
RLsaccess ==> No      Yes | No
LSRPOOLId : 1      1-8 | None
LSRPOOLNum ==> 001    1-255 | None
READInteg ==> Uncommitted Uncommitted | Consistent | Repeatable
DSNSharing ==> Allreqs Allreqs | Modifyreqs
STRings   ==> 001    1-255
Nsrgroup  ==>
REMOTE ATTRIBUTES
REMOTESystem ==>
REMOTENAME ==>
REMOTE AND CFDATATABLE PARAMETERS

SYSID=ESK0 APPLID=ES1ACICS
DEFINE SUCCESSFUL DSN=ITCICSA.CICS560.DFHCSD
PF 1 HELP 2 COM 3 END 6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF 12 CNCL

```

THEN PF8 - FORWARD

ENTER RECORD SIZE AND KEY LENGTH

The parameters for your VSAM file TSOCP01.ESK0G.SKILLS (SKILLFL) are:

RECORDSIZE: 240 240 (Fixed record length of 240 bytes)

KEYS: 4 0 (Key length is 4 bytes, starting at offset 0)

```
OVERTYPE TO MODIFY OR PRESS ENTER TO EXECUTE                CICS RELEASE = 0730
CEDA  DEFIne File( SKILLS  )
+  LSRPOOLNum    ==> 001                      1-255 | None
    READInteg    ==> Uncommitted              Uncommitted | Consistent | Repeatable
    DSNSharing   ==> Allreqs                  Allreqs | Modifyreqs
    STRings      ==> 001                      1-255
    Nsrgroup     ==>
    REMOTE ATTRIBUTES
    REMOTESystem ==>
    REMOTENAME   ==>
    REMOTE AND CFDATATABLE PARAMETERS
    RECORDSize   ==> 0240                      1-32767
    Keylength     ==> 004                      1-255 (1-16 For CF Datatable)
    INITIAL STATUS
    STATUS        ==> Enabled                  Enabled | Disabled | Unenabled
    Opentime      ==> Firstref                 Firstref | Startup
    Disposition   ==> Share                   Share | Old
    BUFFERS
+  Databuffers    ==> 00002                    2-32767

                                                                    SYSID=ESK0 APPLID=ES1ACICS
                                                                    DSN=ITCICSA.CICS560.DFHCS
PF 1  HELP 2  COM 3  END                      6  CRSR 7  SBH 8  SFH 9  MSG 10 SB 11 SF 12 CNCL
```

```

OVERTYPE TO MODIFY OR PRESS ENTER TO EXECUTE                CICS RELEASE = 0730
CEDA DEFINE File( SKILLS )
+ RECORDSize    ==> 00240                1-32767
  Keylength     ==> 004                  1-255 (1-16 For CF Datatable)
INITIAL STATUS
  Status        ==> Enabled              Enabled | Disabled | Unenabled
  Opentime      ==> Firstref             Firstref | Startup
  Disposition   ==> Share                Share | Old
BUFFERS
  Databuffers   ==> 00002                2-32767
  Indexbuffers  ==> 00001                1-32767
DATATABLE PARAMETERS
  TABLE        ==> No                  No | Cics | User | CF
  Maxnumrecs    ==> Nolimit              Nolimit | 1-99999999
CFDATATABLE PARAMETERS
  CFdtpool      ==>
  TABLName      ==>
  UPDATEModel   ==> Locking              Contention | Locking
+ Load          ==> No                  No | Yes

                                                                    SYSID=ESK0 APPLID=ES1ACICS
                                                                    DSN=ITCICSA.CICS560.DFHCS
PF 1 HELP 2 COM 3 END                6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF 12 CNCL

```

PAGE DOWN AGAIN AND ENTER YES ON ALL THE OPERATIONS

OVERTYPE TO MODIFY OR PRESS ENTER TO EXECUTE

CICS RELEASE = 0730

CEDA DEFINE File(SKILLS)

+ Indexbuffers ==> 00001 1-32767

DATATABLE PARAMETERS

TABLE	==> No	No CICS User CF
Maxnumrecs	==> Nolimit	Nolimit 1-99999999

CFDATATABLE PARAMETERS

CFdtpool	==>	
TABLEName	==>	
UPDATEModel	==> Locking	Contention Locking
Load	==> No	No Yes

DATA FORMAT

RECORDFormat	==> V	V F
--------------	-------	-------

OPERATIONS

Add	==> YES	No Yes
BRowse	==> YES	No Yes
DELeTe	==> YES	No Yes
READ	==> Yes	Yes No
+ UPDATE	==> YES	No Yes

SYSID=ESK0 APPLID=ES1ACICS

DSN=ITCICSA CICS560 DEHCSD

ENTER

ENTER

```

OVERTYPE TO MODIFY                                CICS RELEASE = 0730
CEDA DEFINE File( SKILLS )
+ Indexbuffers ==> 00001                          1-32767
DATATABLE PARAMETERS
TABLE ==> No                                       No | Cics | User | CF
Maxnumrecs ==> Nolimit                          Nolimit | 1-99999999
CFDATATABLE PARAMETERS
CFdtpool ==>
TABLEName ==>
UPDATEModel ==> Locking                         Contention | Locking
Load ==> No                                       No | Yes
DATA FORMAT
RECORDFormat ==> V                               V | F
OPERATIONS
Add ==> Yes                                       No | Yes
BRowse ==> Yes                                   No | Yes
DELeTe ==> Yes                                   No | Yes
READ ==> Yes                                     Yes | No
+ UPDATE ==> Yes                                 No | Yes

                                           SYSID=ESK0 APPLID=ES1ACICS
DEFINE SUCCESSFUL                               DSN=ITCICSA.CICS560.DFHCS
PF 1 HELP 2 COM 3 END                          6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF 12 CNCL

```

DEFINE SUCCESSFUL

CEDA INS FILE(SKILLS) G(ESK0G)

```
INS FILE(SKILLS) G(ESK0G)
OVERTYPE TO MODIFY
CEDA  Install
  ATomservice  ==> 
  Bundle       ==> 
  CONnection   ==> 
  CORbaserver  ==> 
  DB2Conn      ==> 
  DB2Entry     ==> 
  DB2Tran      ==> 
  DJar         ==> 
  DOctemplate  ==> 
  DUmppcode    ==> 
  Enqmodel     ==> 
  File         ==> SKILLS
  Ipconn       ==> 
  JOurnalmodel ==> 
  JVmserver    ==> 
  LIBrary      ==> 
+ LSRpool      ==> 

                                SYSID=ESK0 APPLID=ES1ACICS
INSTALL SUCCESSFUL             DSN=ITCICSA.CICS560.DFHCSD
PF 1 HELP                      3 END                      6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF 12 CNCL
                                04/022
```

SEE FAILED MESSAGE ,. IT ALREADY EXIST AND THAT IS OK

PF3

CHECK EXISTING FILE:

CEMT I FILE(SKILLS)

NOTE: IN THE PROGRAM, i CAN GET THE NICK NAME THEN COME HERE TO GET THE ACTUAL FILE NAME

```
I FILE(SKILLS) ■
STATUS: RESULTS - OVERTYPE TO MODIFY
Fil(SKILLS ) Vsa Clo Ena Rea Upd Add Bro Del      Sha
          Dsn( TS0CP01.ESK0G.SKILLS )
```

fyi

IF THERE IS A MISTAKE AND YOU WANT TO DISCARD THIS FILE AND ASSOCIATE A DIFFERENT FILE TO THE NICKNAME

HIT PF3

CHECK IF THE FILE IS DISABLED

CEMT DISC FILE(SKILLS)

```
DISC FILE(EMPKSDS)
STATUS:  RESULTS
Fil(EMPKSDS )
```

NOT DISABLED

Now, disable the file

CEMT I FILE(SKILLS)

```
I FILE(EMPKSDS) -  
STATUS: RESULTS - OVERTYPE TO MODIFY  
Fil(EMPKSDS ) Vsa Clo Ena Rea Upd Add Bro Del Sha  
Dsn( S1304.CICS.EMPDETAL )
```

Change Ena to Dis

```
I FILE(EMPKSDS) -  
STATUS: RESULTS - OVERTYPE TO MODIFY  
Fil(EMPKSDS ) Vsa Clo Dis Rea Upd Add Bro Del Sha NORMAL  
Dsn( S1304.CICS.EMPDETAL )
```

Now discard

CEMT DISC FILE(SKILLS)

```
DISC FILE(EMPKSDS )
STATUS: RESULTS
Fil(EMPKSDS )
```

RESOURCE DISCARDED

Now can Reinstall
CEDA INS FILE(SKILLS) G(ESK0G)

Now recheck the associated file

CEMT I FILE(SKILLS)

I FILE(EMPKSDS)

STATUS: RESULTS - OVERTYPE TO MODIFY

File(EMPKSDS) Vsa Clo Ena Rea Upd Add Bro Del Sha

Dsn(S1304.CICS.KSDS)

tESTING IN cisS

1. Make sure everything is connecting using tranid : MFTR Error not found	Type: CEDF ,OFF Typ: MYTR Error: Program Not available
--	--

mytr

DFHAC2016 09/05/2026 16:03:17 ES1ACICS Transaction MYTR cannot run because
program MFRNMAPP is not available.

CEMT I PROG(MFRNMAPP)

INQUIRE PROGRAM(MFRNMAPP) ■

STATUS: RESULTS - OVERTYPE TO MODIFY

Prog(MFRNMAPP)	Leng(0000000000)	Map	Ena	Pri	Req
Resc(0000)	Use(0000000000)			Qua	Cic

Oops looks like I defined the program as a mapset.

```
CEDA DEFINE PROGRAM(MFRNMAPP) GROUP(ESK0G)  
  LANGUAGE(COBOL)
```

```
CEda def program(mfrnmapp) group(esk0g) language(cobol) ■  
STATUS:  SESSION ENDED
```

```

DEF PROGRAM(MFRNMAPP) GROUP(ESK0G) LANGUAGE(COBOL)
OVERTYPE TO MODIFY                                     CICS RELEASE
CEDA Define PROGram( MFRNMAPP )
  PROGram      : MFRNMAPP
  Group        : ESK0G
  DEScription  ==> 
  Language     ==> CObol      CObol | Assembler | Le370 | C |
  REload       ==> No        No | Yes
  RESident     ==> No        No | Yes
  USAge        ==> Normal    Normal | Transient
  USElpacopy   ==> No        No | Yes
  Status       ==> Enabled   Enabled | Disabled
  RSl          : 00         0-24 | Public
  CEdf         ==> Yes       Yes | No
  DAtalocation ==> Any       Below | Any
  EXECKey      ==> User      User | Cics
  COncurrency  ==> Quasirent Quasirent | Threadsafe | Require
  Api          ==> Cicsapi   Cicsapi | Openapi
  REMOTE ATTRIBUTES
+  DYnamic      ==> No       No | Yes

                                           SYSID=ESK0 APPLID=
  DEFINE SUCCESSFUL                      DSN=ITCICSA.CICS560.DFHCS
PF 1 HELP 2 COM 3 END                    6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF

```

PF3

CEDA INS PROGRAM(MFRNMAPP) GROUP(ESK0G)

INS PROGRAM(MFRNMAPP) GROUP(ESK0G)

OVERTYPE TO MODIFY

CEDA Install

ATomservice	==>	
Bundle	==>	
CONnection	==>	
CORbaserver	==>	
DB2Conn	==>	
DB2Entry	==>	
DB2Tran	==>	
DJar	==>	
D0ctemplate	==>	
DUmpcode	==>	
Enqmodel	==>	
File	==>	
Ipconn	==>	
J0urnalmodel	==>	
JVmserver	==>	
LIBrary	==>	
+ LSrpool	==>	

INSTALL SUCCESSFUL

DSN=ITCICSA.CICS560.DFHCS

PF 1 HELP

3 END

6 CRSR 7 SBH 8 SFH 9 M

Double check all the resources

Cemt DI G(ESK0G)

```
DI G(ESK0G)
ENTER COMMANDS
NAME      TYPE      GROUP      LAST
SKILLS    FILE      ESK0G      09/05/1
MFRNMAPP  MAPSET    ESK0G      09/05/1
MFRNSET   MAPSET    ESK0G      09/05/1
MFRNMAPP  PROGRAM   ESK0G      09/05/1
MYTR      TRANSACTION ESK0G      09/05/1
```

DELETE the MAPSET: MFRNMAPP

DI G(ESKOG)

ENTER COMMANDS

NAME	TYPE	GROUP		LAST CHANGE
SKILLS	FILE	ESKOG		09/05/26 15:46:23
MFRNMAPP	MAPSET	ESKOG	del _	09/05/26 15:33:27
MFRNSET	MAPSET	ESKOG		09/05/26 15:31:51
MFRNMAPP	PROGRAM	ESKOG		09/05/26 16:09:56
MYTR	TRANSACTION	ESKOG		09/05/26 15:36:42

DI G(ESKOG)

ENTER COMMANDS

NAME	TYPE	GROUP	
SKILLS	FILE	ESKOG	
			*el _
MFRNSET	MAPSET	ESKOG	
MFRNMAPP	PROGRAM	ESKOG	
MYTR	TRANSACTION	ESKOG	

Pf3

Pf3

DI G(ESKOG)

ENTER COMMANDS

NAME	TYPE	GROUP
SKILLS	FILE	ESKOG
MFRNSET	MAPSET	ESKOG
MFRNMAPP	PROGRAM	ESKOG
MYTR	TRANSACTION	ESKOG

Type: Typ: MYTR

Message: mfrnmapp is not available

```
mytr  
_____  
  
DFHAC2016 09/05/2026 16:21:53 ES1ACICS Transaction MYTR cannot run because  
program MFRNMAPP is not available.
```

```
I PROG(MFRNMAPP)
STATUS: RESULTS - OVERTYPE TO MODIFY
  Prog(MFRNMAPP) Leng(0000000000) Cob Pro Ena Pri Req Ced
    Resc(0000) Use(0000000000) Any Uex Ful Qua Cic
```

1.Issue NEWCOPY:Force CICS Load Attempt.

Overtime I PROG(MFRNMAPP) on your current screen with:

CEMT SET PROGRAM(MFRNMAPP) NEWCOPY

and press Enter.

```
CEMT set program(mfrnmapp) newcopy_
STATUS: SESSION ENDED
```

```
SET PROGRAM(MFRNMAPP) NEWCOPY
STATUS: RESULTS - OVERTYPE TO MODIFY
  Prog(MFRNMAPP) Len(0000000000) Cob Pro Ena Pri Req Ced      LOAD FAILED
    Res(0000) Use(0000000000) Any Uex Ful Qua Cic
```


To fix:

Resubmitted my JCL TSOCP01.ESK0G.JCL(MYCICJCL)

```
//TSOCP01X JOB 30000000,'ESK0 JOB CARD ',MSGLEVEL=(1,1),
//  CLASS=A,MSGCLASS=Q,NOTIFY=&SYSUID,REGION=0M
//*****
//*
//*****
//TRN      EXEC PGM=DFHECP1$,REGION=0M,
//          PARM=(CICS)
//STEPLIB  DD DSN=CICS560.CICS.SDFHLOAD,DISP=SHR
//SYSPRINT DD SYSOUT=*
//SYSPUNCH DD DSN=&&SYSCIN,DISP=(,PASS),
//          SPACE=(CYL,(1,1)),
//          DCB=(RECFM=FB,LRECL=80,BLKSIZE=400)
//SYSIN    DD DISP=SHR,DSN=TSOCP01.ESK0G.PROGRAMS(MFRNMAPP)
//*----- EXEC the COBOL II Compiler. -----
//COB      EXEC PGM=IGYCRCTL,REGION=0M,
//          COND=(7,LT),
//          PARM=(APOST,NODYNAM,OPTIMIZE,SSRANGE,
//          LIST,MAP,XREF,'NOCOMPILE(E)',LIB,RENT)
//STEPLIB  DD DSN=IGY.SIGYCOMP,DISP=SHR
//SYSPRINT DD SYSOUT=*
//SYSLIB   DD DSN=CICS560.CICS.SDFHCOB,DISP=SHR
//          DD DSN=CICS560.CICS.SDFHSAMP,DISP=SHR
//          DD DSN=TSOCP01.ESK0G.COPYLIB,DISP=SHR
```

```

/*----- LinkEdit the output into the specified Load Librar
//LKED      EXEC PGM=IEWL,REGION=2048K,
//          COND=(7,LT),
//          PARM=(LIST,XREF)
//SYSPRINT DD SYSOUT=*
//SYSLIB    DD DSN=CICS560.CICS.SDFHLOAD,DISP=SHR
//          DD DSN=CEE.SCEELKED,DISP=SHR
//SYSUT1    DD SPACE=(CYL,(1,1))
//SYSLIN    DD DSN=CICS560.CICS.SDFHCOB(DFHEILIC),DISP=SHR,
//          DCB=BLKSIZE=400
//          DD DSN=&&OBJMOD,DISP=(OLD,PASS)
//          DD DDNAME=SYSIN
//SYSLMOD   DD DISP=SHR,DSN=TSOCP01.ESK0G.LOADLIB
//SYSIN     DD *
            NAME PAYPGMM(R)
/*
/*----- Delete any temporary datasets -----
//DELTEMP   EXEC PGM=IEFBR14
//DD1       DD DSN=&&SYSCIN,DISP=(OLD,DELETE)
//DD2       DD DSN=&&OBJMOD,DISP=(OLD,DELETE)

```

JOB 1980

```

IGD103I SMS ALLOCATED TO DDNAME SYSIN
IGD101I SMS ALLOCATED TO DDNAME (SYSLIN  )
        DSN (SYS26248.T163240.RA000.TSOCP01X.OBJMOD.H01  )
        STORCLAS (BASE) MGMTCLAS (          ) DATACLAS (          )
        VOL SER NOS= SMS001
IEF142I TSOCP01X COB - STEP WAS EXECUTED - COND CODE 0012
IEF285I IGY.SIGYCOMP                                KEPT
IEF285I VOL SER NOS= ZOS0X1.
IEF285I TSOCP01.TSOCP01X.JOB01980.D0000104.?          SYSOUT
IEF285I CICS560.CICS.SDFHCOB                          KEPT
IEF285I VOL SER NOS= ZOS0CA.
IEF285I CICS560.CICS.SDFHSAMP                          KEPT
IEF285I VOL SER NOS= ZOS001.

```

Noticed

NAME: PAYPGMM(R) SHOULD BE UPDATED

```

44  /*----- LINKED: the output into the specified Load Library -----*/
45  //LKED      EXEC PGM=IEWL,REGION=2048K,
46  //          COND=(7,LT),
47  //          PARM=(LIST,XREF)
48  //SYSPRINT DD SYSOUT=*
49  //SYSLIB    DD DSN=CICS560.CICS.SDFHLOAD,DISP=SHR
50  //          DD DSN=CEE.SCEELKED,DISP=SHR
51  //SYSUT1    DD SPACE=(CYL,(1,1))
52  //SYSLIN    DD DSN=CICS560.CICS.SDFHCOB(DFHEILIC),DISP=SHR,
53  //          DCB=BLKSIZE=400
54  //          DD DSN=&&OBJMOD,DISP=(OLD,PASS)
55  //          DD DDNAME=SYSIN
56  //SYSLMOD   DD DISP=SHR,DSN=TSOCP01.ESK0G.LOADLIB
57  //SYSIN     DD *
58  |      NAME PAYPGMM(R)
59  /*|
60  /*----- Delete any temporary datasets -----*/
61  //DELTEMP   EXEC PGM=IEFBR14
62  //DD1       DD DSN=&&SYSCIN,DISP=(OLD,DELETE)
63  //DD2       DD DSN=&&OBJMOD,DISP=(OLD,DELETE)
64
65

```

UPDATED

```

//          DD DSN=TSOCP01X.JOB01974 - CC 0000
//          DD DDNAME=SYSIN
//SYSLMOD  DD DISP=SHR,DSN=TSOCP01.ESK0G.LOADLIB
✓ //SYSIN  DD *
      NAME MFRNMAPP(R)
/*
//*----- Delete any temporary datasets -----
//DELTEMP EXEC PGM=IEFBR14

```

Reran job: 1981

```

✓ JOBS
  ✓ TSOCP01X(JOB01974) - CC 0000
  > INST000(JOB01977) - CC 0000
  > TSOCP01X(JOB01979) - CC 0012
  > TSOCP01X(JOB01980) - CC 0012
  ✓ TSOCP01X(JOB01981) - CC 0012
    JES2:JESMSG LG(2)
    JES2:JESJCL(3)
    JES2:JESYSMSG(4)
    TRN:SYS PRINT(103)
    COB:SYS PRINT(104)
  > TSOCP01(TSU01976) - ABEND S222

```

Job:

Scroll all the way down

```
End of Data Division Map
1PP 5655-EC6 IBM Enterprise COBOL for z/OS 6.3.0 P191120      MFRNMAPP Date 09/05/2026 Time 16:3
0LineID  Message code  Message text

    IGYSC3002-I  A severe error was found in the program.  The "OPTIMIZE" and the "STGOPT" co
536  IGYP52121-S  "MFRNMAPO" was not defined as a data-name.  The statement was discarded.

    Same message on line:  1076  1080  1089  1093

1025  IGYP52121-S  "MFRNMAPI" was not defined as a data-name.  The statement was discarded.
-Messages      Total      Informational      Warning      Error      Severe      Terminating
0Printed:      9          3                                6
-* Statistics for COBOL program MFRNMAPP:
*   Source records = 1095
*   Data Division statements = 344
*   Procedure Division statements = 267
*   Generated COBOL statements = 0
```

EMS 92 OUTPUT DEBUG CONSOLE TERMINAL IBM Z Open Editor

```
MFRNMAP  DFHMSD TYPE=MAP,MODE=INOUT,EXTATT=YES,LANG=COBOL,TIOAPFX=YES
* --- Screen Header ---
MFRNSET  DFHMDI SIZE=(24,80)
         DFHMDI POS=(1,26),LENGTH=28,ATTRB=(BRT,PROT),
         INITIAL='SKILLS && DETAILS INQUIRY'
```

Fined map name and map set

Checked copy book and only MFRNMAPI was recreated not the out

```
01 MFRNMAPI.  
02 FILLER PIC X(12).  
02 MFDATL COMP PIC S9(4).  
02 MFDATF PICTURE X.  
02 FILLER REDEFINES MFDATF.  
03 MFDATA PICTURE X.  
02 FILLER PICTURE X(4).  
02 MFDATI PIC X(10).  
02 MFTIME COMP PIC S9(4).  
02 MFTIMEF PICTURE X.  
02 FILLER REDEFINES MFTIMEF.  
03 MFTIMEA PICTURE X.  
02 FILLER PICTURE X(4).  
02 MFTIMEI PIC X(8).  
02 MFYEARL COMP PIC S9(4).  
02 MFYEARF PICTURE X.
```

I added mod= INOUT as the definition

FIX:

Need to cpy in the set name and not the map name into the program

```
WORKING-STORAGE SECTION.  
*--- BMS MAP COPYBOOK ---  
COPY MFRNSET.  
      ┘  
*--- DFHAID & DFHBMSCA COPYBOOKS ---  
COPY DFHAID.  
COPY DFHBMSCA.  |
```

Update the JCL for the MAP


```

//INST000 JOB , 'MFRNMAP', REGION=2M,
//          CLASS=A, NOTIFY=&SYSUID, MSGCLASS=A
//*JOBPARM SYSAFF=*
//*****
//*
//* THIS JOB ASSEMBLES AND LINKS AN ASSEMBLER PROGRAM
//*
//*****
//WHERE SMA JCLLIB ORDER='CICS560.CICS.SDFHPROC'
//SETMAP SET MAPNAME=MFRNMAP <<<<< PUT YOUR MAPNAME IN HERE
//MAPASM EXEC PROC=DFHMAPT, MAPNAME=MFRNMAP,
//          INDEX='CICS560.CICS',
//          MAPLIB='TSOCP01.ESK0G.LOADLIB',
//          DSCTLIB='TSOCP01.ESK0G.COPYLIB',
//          TEMPLIB='TSOCP01.ESK0G.TEMPLATE'
//COPY.SYSUT1 DD DSN=TSOCP01.ESK0G.PROGRAMS(MFRNMAP), DISP=SHR

```

Its incorrect also.

**FIXED CODE
TO USE SET NAME**

```

P01.ESK0GJCL(MYCICJCL).jcl • = TSOCP01.ESK0GJCL(MYASMJCL).jcl • Welcome to Z Open
c: > Users > IBM > .vscode-oss > extensions > zowe.vscode-extension-for-zowe-1.5.1 > resources > t
1  //INST000 JOB , 'MFRNMAP', REGION=2M,
2  //          CLASS=A, NOTIFY=&SYSUID, MSGCLASS=A
3  //*JOBPARM SYSAFF=*
4  //*****
5  //*
6  //* THIS JOB ASSEMBLES AND LINKS AN ASSEMBLER PROGRAM
7  //*
8  //*****
9  //WHERE SMA JCLLIB ORDER='CICS560.CICS.SDFHPROC'
10 //SETMAP SET MAPNAME=MFRNSET <<<<< PUT YOUR MAPNAME IN HERE
11 //MAPASM EXEC PROC=DFHMAPT, MAPNAME=MFRNSET,
12 //      INDEX='CICS560.CICS',
13 //      MAPLIB='TSOCP01.ESK0G.LOADLIB',
14 //      DSCTLIB='TSOCP01.ESK0G.COPYLIB',
15 //      TEMPLIB='TSOCP01.ESK0G.TEMPLATE'
16 //COPY.SYSUT1 DD DSN=TSOCP01.ESK0G.PROGRAMS(MFRNMAP), DISP=SHR
17

```

Correct order

```

c: > Users > IBM > .vscode-oss > extensions > zowe.vscode-extension-for-zowe-1.5.1 > resources > temp > _D_
1  MFRNSET  DFHMSD TYPE=MAP,MODE=INOUT,EXTATT=YES,LANG=COBOL,TIOAPFX=YES
2  * --- Screen Header ---
3  MFRNMAP  DFHMDI SIZE=(24,80)
4  DFHMDF POS=(1,26),LENGTH=28,ATTRB=(BRT,PROT),
5  INITIAL='SKILLS && DETAILS INQUIRY'
6  * --- Date & Time ---
7  DFHMDF POS=(1,58),LENGTH=5,ATTRB=(NORM,PROT),INITIAL='DATE:'
8  MFRNMAP  DFHMDF POS=(1,64),LENGTH=10,ATTRB=(BRT,PROT)

```

CORRECTED PROGRAM: TO USE THE set name

```

AUTHOR: TIONA.
*=====*
* FULL CRUD CICS PROGRAM MATCHING MFRNMAP *
*=====*
ENVIRONMENT DIVISION.
DATA DIVISION.

WORKING-STORAGE SECTION.
*--- BMS MAP COPYBOOK ---
COPY MFRNSET.

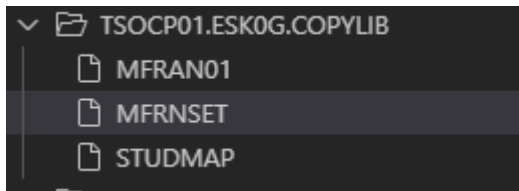
*--- DFHAID & DFHBMSCA COPYBOOKS ---
COPY DFHAID.
COPY DFHBMSCA.

*--- VSAM RECORD STRUCTURE (MATCHES RECORD LENGTH = 240 BYTES) ---
01 WS-SKILL-DETAIL-RECORD.

```

I NEED TO CHAGNE MY **MAP PROGRAM NAME** TO THE SET NAME
BECAUSE PROGRAM NAME IS COMING OUT AS THE COPYBOOK NAME I
will use in my program

DONE



Copybook has both MFRNMAPI and MFRNMAPO

So will now rerun the program

Job 1986 CC0000

RECOMPILE PROGRAM USING THE COPYBOOKS

back to CICS

CEMT DEF MAPSET(MFRNSET) GROUP(ESK0G)

```

DEFINE MAPSET(MFRNSET) GROUP(ESK0G)
OVERTYPE TO MODIFY
CEDA DEFINE MAPset( MFRNSET )
  MAPset      : MFRNSET
  Group       : ESK0G
  DESCRIPTION ==>
  RESident    ==> No           No | Yes
  USAge       ==> Normal      Normal | Transient
  USElpacopy  ==> No           No | Yes
  Status      ==> Enabled     Enabled | Disabled
  RSL         : 00           0-24 | Public
DEFINITION SIGNATURE
  DEFinetime  : 09/05/26 15:31:51
  CHANGETime  : 09/05/26 15:31:51
  CHANGEUsrid : CICSSTC
  CHANGEAGEnt : CSDApi       CSDApi | CSDBatch
  CHANGEAGRel : 0730

S Object already exists in this group.
                                SYSID=ESK0 APPLID=ES1ACICS
DEFINE UNSUCCESSFUL           DSN=ITCICSA.CICS560.DFHCSO
F 1 HELP 2 COM 3 END          6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF 12 CNCL

```

CEMT INS MAPSET(MFRNSET) GROUP(ESK0G)

Check the PROGRAM Definition: Type **CEDA INQUIRE PROGRAM(MFRNMAPP)**
GROUP(ESK0G)

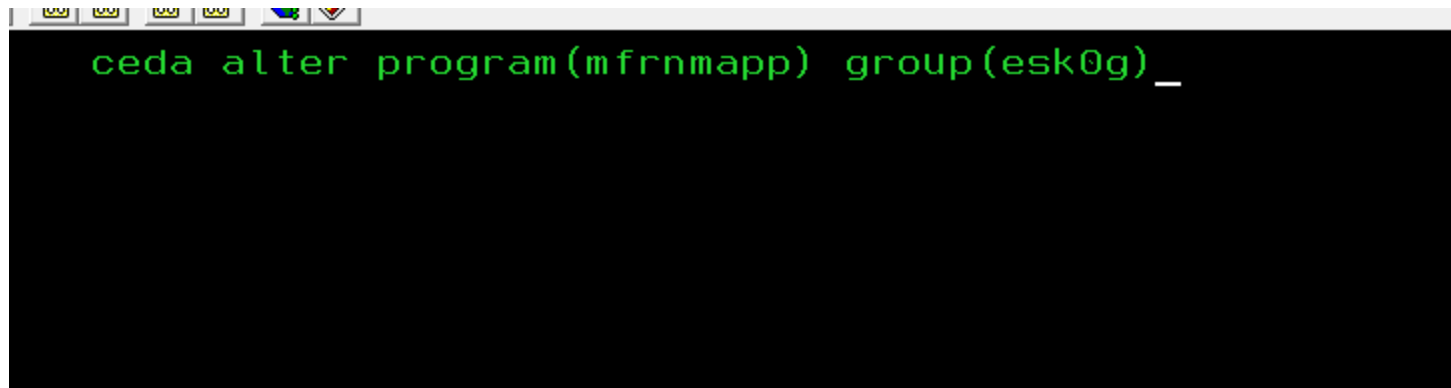
```

I PROGRAM(MFRNMAPP)
OVERTYPE TO MODIFY
CEDA Install
  ATomservice ==> -
  Bundle ==>
  CONnection ==>
  CORbaserver ==>
  DB2Conn ==>
  DB2Entry ==>
  DB2Tran ==>
  DJar ==>
  D0ctemplate ==>
  DUmrcode ==>
  Enqmodel ==>
  File ==>
  Ipconn ==>
  JOurnalmodel ==>
  JVmserver ==>
  LIBrary ==>
+ LSrpool ==>
S GROUP or LIST must be specified.

                                SYSID=ESK0 APPLID=ES1ACICS
                                DSN=ITCICSA.CICS560.DFHCSO
PF 1 HELP          3 END          6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF 12 CNCL
                                04/022

```

Ensure Language is set to COBOL and Status is ENABLED.

A terminal window with a light gray title bar containing several icons. The main area is black with green text. The text is a command: 'ceda alter program(mfrnmapp) group(esk0g)_' followed by a cursor.

```
ceda alter program(mfrnmapp) group(esk0g)_
```

Confirmed

```

ALTER PROGRAM(MFRNMAPP) GROUP(ESK0G)
OVERTYPE TO MODIFY
CEDA ALter PROGram( MFRNMAPP )
  PROGram          : MFRNMAPP
  Group            : ESK0G
  DEScription      ==>
  Language         ==> CObol          CO
  RELoad           ==> No             No
  RESident         ==> No             No
  USAge            ==> Normal         No
  USElpacopy       ==> No             No
  Status           ==> Enabled       En
  RSl              : 00              0-
  CEdf             ==> Yes            Yes
  Datalocation     ==> Any            Be
  EXECKey          ==> User           Us
  COncurrency      ==> Quasirent     Qu
  Api              ==> Cicsapi        Ci
  REMOTE ATTRIBUTES
+  DYnamic         ==> No            No

      ALTER SUCCESSFUL                DSN=ITCICS
PF 1 HELP 2 COM 3 END                6 CRSR

```

Video capture

Enter data - Press PF2	Record written successfully
Enter data - Press PF2	Record written successfully
PF3 to clear screen	
Cedf	Debug mode
First time	Get time in Hexadecimal format Format time: format the date and Send empty map
Second time	

Check the TRANSACTION Definition: Type **CEDA INQUIRE TRANSACTION(MYTR) GROUP(ESK0G)** and press **Enter**.

- Ensure **Program** is set to **MFRNMAPP**.

```
I TRANS(MYTR) GROUP(ESK0G)
```

```
OVERTYPE TO MODIFY
```

```
CEDA Install
```

```
ATomservice ==> -
```

```
Bundle ==>
```

```
CONnection ==>
```

```
CORbaserver ==>
```

```
DB2Conn ==>
```

```
DB2Entry ==>
```

```
DB2Tran ==>
```

```
DJar ==>
```

```
D0ctemplate ==>
```

```
DUmpcode ==>
```

```
Enqmodel ==>
```

```
File ==>
```

```
Ipconn ==>
```

```
J0urnalmodel ==>
```

```
JVmserver ==>
```

```
LIBrary ==>
```

```
+ LSrpool ==>
```

```
SYSID=ESK0 APPLID=ES1ACICS
```

```
INSTALL SUCCESSFUL
```

```
DSN=ITCICSA.CICS560.DFHCS
```

```
PF 1 HELP
```

```
3 END
```

```
6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF 12 CNCL
```

Currently installed the transaction MYTR, but you still need to tie it to your program MFRNMAPP or update the transaction definition so CICS knows which program to execute when you run MYTR.

```
CEDA DEF PROGRAM(MFRNMAPP) GROUP(ESK0G) LANGUAGE(COBOL) STATUS(ENABLE) _  
STATUS:  SESSION ENDED
```

aLREADY EXIST

```
DEF PROGRAM(MFRNMAPP) GROUP(ESK0G) LANGUAGE(COBOL) STATUS(ENABLE)
OVERTYPE TO MODIFY CICS RELEASE = 0730
```

```
CEDA DEFine PROGram( MFRNMAPP )
```

```
  PROGram      : MFRNMAPP
```

```
  Group        : ESK0G
```

```
  DEScription  ==> _
```

```
  Language     ==> CObol      CObol | Assembler | Le370 | C | Pli
```

```
  REload       ==> No         No | Yes
```

```
  RESident     ==> No         No | Yes
```

```
  USAge        ==> Normal | Normal | Transient
```

```
  USElpacopy   ==> No         No | Yes
```

```
  Status       ==> Enabled   Enabled | Disabled
```

```
  RSl          : 00         0-24 | Public
```

```
  CEdf         ==> Yes       Yes | No
```

```
  DAtalocation ==> Any       Below | Any
```

```
  EXECKey      ==> User      User | Cics
```

```
  COncurrency  ==> Quasirent Quasirent | Threadsafe | Required
```

```
  Api          ==> Cicsapi   Cicsapi | Openapi
```

```
REMOTE ATTRIBUTES
```

```
+ DYnamic      ==> No         No | Yes
```

```
S Object already exists in this group.
```

```
SYSID=ESK0 APPLID=ES1ACICS
```

```
DEFINE UNSUCCESSFUL
```

```
DSN=ITCICSA.CICS560.DFHCSD
```

```
PF 1 HELP 2 COM 3 END
```

```
6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF 12 CNCL
```

```
06/022
```

```
CEDA DEF TRANS(MYTR) GROUP(ESKOG) PROGRAM(MFRNMAPP) _  
STATUS:  SESSION ENDED
```

```

DEF TRANS(MYTR) GROUP(ESK0G) PROGRAM(MFRNMAPP)
OVERTYPE TO MODIFY
CEDA DEFINE TRANSAction( MYTR )
  TRANSAction      : MYTR
  Group            : ESK0G
  DEScription      ==>
  PROGram          ==> MFRNMAPP
  TWAsize          ==> 00000          0-32767
  PROFile          ==> DFHCICST
  PArtitionset     ==>
  STatus           ==> Enabled        Enabled | Disabled
  PRIMedsize       : 00000          0-65520
  TASKDATALoc      ==> Any           Below | Any
  TASKDATAKey      ==> User          User | Cics
  STOrageclear     ==> No            No | Yes
  RUNaway          ==> System        System | 0 | 250-2700000
  SHutdown         ==> Disabled      Disabled | Enabled
  ISolate          ==> Yes           Yes | No
  Brexit           ==>
+ REMOTE ATTRIBUTES
  S Object already exists in this group.
                                           SYSID=ESK0 APPLID=ES1ACICS
  DEFINE UNSUCCESSFUL                DSN=ITCICSA.CICS560.DFHCSD
PF 1 HELP 2 COM 3 END                6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF 12 CNCL
                                           06/022

```

e server/host pokedvm2.ilsvpn.ibm.com using port 9023

1. Check Program Load Status in CEMT: 1 min.

Clear your CICS screen (**Ctrl + R**), type the following command, and press **Enter**:

CEMT INQUIRE PROGRAM(MFRNMAPP)

How to verify: Look at the output line to see if it shows Ena (Enabled), Cob (COBOL), and the loaded length (e.g. Leng(00004096)).

```
I PROGRAM(MFRNMAPP)
STATUS: RESULTS - OVERTYPE TO MODIFY
  Prog(MFRNMAPP) Leng(0000000000) Cob Pro Ena Pri Req Ced
    Resc(0000) Use(0000000000) Any Uex Ful Qua Cic
```

UNABLE TO LOCATE LOAD MODULE

WILL SEE IF I CAN USE THE D62WW:SYSLMOD
JOB 1988

```
54 //      DD DSN=&&OBJMOD,DISP=(OLD,PASS)
55 //      DD DDNAME=SYSIN
56 //SYSLMOD DD DISP=SHR,DSN=D62WW.ESK0G.LOADLIB
57 //SYSIN DD *
58 |     NAME MFRNMAPP(R)
59 /*
60 /*----- Delete any temporary datasets -----
61 //DELTEMP EXEC PGM=IEFBR14
62 //DD1      DD DSN=&&SYSCIN,DISP=(OLD,DELETE)
63 //DD2      DD DSN=&&OBJMOD,DISP=(OLD,DELETE)
64
65
```

If **Leng** is still **0**, force CICS to re-read the library by typing **NEW** directly over **Ena** on that line and pressing Enter, or run: CEMT SET PROGRAM(MFRNMAPP) NEWCOPY


```
CEMT SET PROGRAM(MFRNMAPP) NEWCOPY
STATUS:  RESULTS - OVERTYPE TO MODIFY
  Prog(MFRNMAPP) Len(0000013400) Cob Pro Ena Pri      Ced      NORMAL
    Res(0000) Use(00000000000) Any Uex Ful Qua Cic
```

cLEARED cics

TYPE: MYTR

ERROR:

```
DFHAC2206 18:08:48 ES1ACICS Transaction MYTR failed with abend APCT. Updates
to local recoverable resources backed out.
```

01/001

Abend **APCT** means CICS tried to load a program or mapset requested by **MYTR**, but **could not find it or it is disabled/not installed**.

Since program **MFRNMAPP** is already loaded, APCT happens when the **MAPSET(MFRNSET)** is missing or not installed in CICS.

NEW LIBRARY

Abend **APCT** means CICS tried to load a program or mapset requested by **MYTR**, but **could not find it or it is disabled/not installed**.

Since program **MFRNMAPP** is already loaded, APCT happens when the **MAPSET(MFRNSET)** is missing or not installed in CICS.

```
CEMT SET MAPSET(MFRNSET) NEWCOPY_
```

LOAD FAILED

2. NEWCOPY MAPSET in CEMT:1 min. Clear the screen (Ctrl + R) and issue: CEMT SET MAPSET(MFRNSET) NEWCOPY

```
SET PROGRAM(MFRNSET) NEWCOPY ■
STATUS: RESULTS - OVERTYPE TO MODIFY
Prog(MFRNSET ) Len(0000000000)      Map Ena Pri Req      LOAD FAILED
      Res(0000) Use(0000000000)      Qua Cic
```

2. Define MAPSET via CEDA:1 min. On the blank screen, type the exact command and press Enter: CEDA DEFINE PROGRAM(MFRNSET) GROUP(ESK0G) LANGUAGE(ASSEMBLER)

```

DEF PROGRAM(MFRNSET) GROUP(ESK0G) LANGUAGE(ASSEMBLER)
OVERTYPE TO MODIFY                                     CICS RELEASE = 0730
CEDA DEFINE PROGram( MFRNSET )
  PROGram      : MFRNSET
  Group        : ESK0G
  DEScription  ==> ■
  Language     ==> Assembler      CObol | Assembler | Le370 | C | Pl1
  RELoad       ==> No                No | Yes
  RESident     ==> No                No | Yes
  USAge        ==> Normal           Normal | Transient
  USElpacopy   ==> No                No | Yes
  Status       ==> Enabled          Enabled | Disabled
  RSl          : 00                0-24 | Public
  CEdf         ==> Yes                Yes | No
  DAtalocation ==> Any                Below | Any
  EXECKey      ==> User              User | Cics
  CONcurrency  ==> Quasirent        Quasirent | Threadsafe | Required
  Api          ==> Cicsapi          Cicsapi | Openapi
  REMOTE ATTRIBUTES
+  DYnamic      ==> No                No | Yes

                                                                    SYSID=ESK0 APPLID=ES1ACICS
  DEFINE SUCCESSFUL          DSN=ITCICSA.CICS560.DFHCS
PF 1 HELP 2 COM 3 END      6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF 12 CNCL
                                                                    06/022

```

3. Install MAPSET into CICS Region:1 min.
On the command line, type and press Enter:

CEDA INSTALL PROGRAM(MFRNSET) GROUP(ESK0G)

```
INS PROGRAM(MFRNSET) GROUP(ESK0G)
OVERTYPE TO MODIFY
CEDA Install
  ATomservice ==>
  Bundle ==>
  CONnection ==>
  CORbaserver ==>
  DB2Conn ==>
  DB2Entry ==>
  DB2Tran ==>
  DJar ==>
  DOctemplate ==>
  DUmrcode ==>
  Enqmodel ==>
  File ==>
  Ipconn ==>
  JOurnalmodel ==>
  JVmserver ==>
  LIBrary ==>
+ LSRpool ==>

                                SYSID=ESK0 APPLID=ES1ACICS
INSTALL SUCCESSFUL             DSN=ITCICSA.CICS560.DFHCSO
PF 1 HELP                      3 END
                                6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF 12 CNCL
```

IOAD FAILED AGAIN

How to verify: Look for INSTALL SUCCESSFUL at the bottom.

4.Issue NEWCOPY in CEMT:1 min.

Clear the screen (Ctrl + R), type the following command, and press Enter:

```
CEMT SET PROGRAM(MFRNSET) NEWCOPY
```

How to verify: Your mapset will change to NORMAL with a positive length like Leng(0000002048)!

```
SET PROGRAM(MFRNSET) NEWCOPY ■
STATUS: RESULTS - OVERTYPE TO MODIFY
Prog(MFRNSET ) Len(0000000000) Ass Pro Ena Pri Ced LOAD FAILED
Res(0000) Use(0000000000) Any Uex Ful Qua Cic
```

Why CICS Uses DFHRPL

- **IBM Default Standard:** Since early CICS releases, IBM designed the region startup JCL to read static application load libraries, mapsets, and system modules from the //DFHRPL DD concatenatio

```
I LIBRARY ■
STATUS: RESULTS - OVERTYPE TO MODIFY
Lib(DFHRPL ) Sea(0001) Ran( 10 ) Num(007) Ena Cri
      Dsname01(SYS2.CICS560.SDFHLOAD )
Lib(PAYLIB ) Sea(0002) Ran( 50 ) Num(001) Ena
      Dsname01(D62WW.ESK0G.LOADLIB )
```

FIX : i RERAN MY PROGRAM JCL SO MY PROGRAM LOAD GOES TO MY HLQ

THEN i WILL CREATE A LIBRARY FOR MY TSOPC01.ESK0G. LOADLIB
JOB1993 - DONE

cREATE MYOWN LIBRARY:

CEMT SET PROGRAM(MFRNMAPP) NEWCOPY

```
SET PROGRAM(MFRNMAPP) NEWCOPY ■
STATUS: RESULTS - OVERTYPE TO MODIFY
Prog(MFRNMAPP) Len(0000013400) Cob Pro Ena Pri      Ced      NORMAL
      Res(0000) Use(00000000001) Any Uex Ful Qua Cic
```


CEDA DEFINE LIBRARY(TSOLIB) GROUP(ESK0G) DSNAME01(TSOCP01.ESK0G.LOADLIB)

```
CEDA DEF LIBRARY(TSOLIB) GROUP(ESK0G) DSNAME01(TSOCP01.ESK0G.LOADLIB) RANKING(10)
■
```

```
CEDA INS LIBRARY(TSOLIB) GROUP(ESK0G) ■
STATUS:  SESSION ENDED
```

INS LIBRARY(TSOLIB) GROUP(ESK0G)

OVERTYPE TO MODIFY

CEDA Install

ATomservice ==> ☒

Bundle ==>

CONnection ==>

CORbaserver ==>

DB2Conn ==>

DB2Entry ==>

DB2Tran ==>

DJar ==> ☐

D0ctemplate ==>

DUmpcode ==>

Enqmodel ==>

File ==>

Ipconn ==>

J0urnalmodel ==>

JVmserver ==>

LIBrary ==> TSOLIB

+ LSrpool ==>

INSTALL SUCCESSFUL

PF 1 HELP 3 END

SYSID=ESK0 APPLID=ES1ACICS

DSN=ITCICSA.CICS560.DFHCSO

6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF 12 CNCL

^

04/022

```

DEF LIBRARY(TSOLIB) GROUP(ESK0G) DSN=ITCICSA.CICS560.DFHCS
OVERTYPE TO MODIFY                                CICS RELEASE = 0730
CEDA Define Library( TSOLIB )
  Library      : TSOLIB
  Group        : ESK0G
  Description   ==> 
  Ranking       ==> 50          1-9 | 11-99
  Critical      ==> No         No | Yes
  Status        ==> Enabled    Enabled | Disabled
DATA SET NAMES
DSNAME01 ==> TSOLIB.LOADLIB
DSNAME02 ==> 
DSNAME03 ==> 
DSNAME04 ==> 
DSNAME05 ==> 
DSNAME06 ==> 
DSNAME07 ==> 
DSNAME08 ==> 
DSNAME09 ==> 
DSNAME10 ==>

                                SYSID=ESK0 APPLID=ES1ACICS
DEFINE SUCCESSFUL              DSN=ITCICSA.CICS560.DFHCS
PF 1 HELP 2 COM 3 END          6 CRSR 7 SBH 8 SFH 9 MSG 10 SB 11 SF 12 CNCL
                                06/022

```

2.Install the LIBRARY into CICS:1 min.

Type this command and press Enter:

CEDA INSTALL LIBRARY(TSOLIB) GROUP(ESK0G)

How to verify: Look for INSTALL SUCCESSFUL at the bottom of the screen.

3.NOW run NEWCOPY on Mapset MFRNSET:1 min.

Clear the screen (**Ctrl + R**) and run:

CEMT SET PROGRAM(MFRNSET) NEWCOPY

```
SET PROGRAM(MFRNSET) NEWCOPY ■
STATUS:  RESULTS - OVERTYPE TO MODIFY
  Prog(MFRNSET ) Len(0000001456) Ass Pro Ena Pri      Ced      NORMAL
    Res(0000) Use(00000000000) Any Uex Ful Qua Cic
```

SEE MY NEW LIBRARY

I LIBRARY ■

STATUS: RESULTS - OVERTYPE TO MODIFY

```
Lib(DFHRPL ) Sea(0001) Ran( 10 ) Num(007) Ena Cri
    Dsname01(SYS2.CICS560.SDFHLOAD
Lib(PAYLIB ) Sea(0002) Ran( 50 ) Num(001) Ena
    Dsname01(D62WW.ESK0G.LOADLIB
Lib(TSOLIB ) Sea(0003) Ran( 50 ) Num(001) Ena
    Dsname01(TSOCP01.ESK0G.LOADLIB
```

TEST TRANID:MYTR

SKILLS & DETAILS INQUIRY

DATE: 2026/09/05

TIME: 18:55:44

Marlainna you still Got it!

ENTER YEAR : CATEGORY : (S=Skills, P=Project

ROLE/TITLE :

SKILLS :

HIGHLIGHTS :

LINK/REF :

Enter Year and press F1 to Read or F2 to Add.
F1=READ F2=WRITE F3=EXIT F4=UPDATE F5=DELETE
F6=CLEAR F7=PREV F8=NEXT F10=CONFIRM UPDATE
ENTER=BROWSE

FIXED MY TRANID: MYTR
RECOMPILED THE PROGRAM JOB 1995

Once you update your transaction definition in **CEDA**:

1.Install Updated Transaction:1 min.

Clear the screen (**Ctrl + R**) and run:

CEDA INSTALL TRANSACTION(MYTR) GROUP(ESK0G)

How to verify: Look for INSTALL SUCCESSFUL at the bottom of the screen.

2.Refresh Mapset and Program:1 min.

Clear screen (**Ctrl + R**) and run:

CEMT SET PROGRAM(MFRNMAPP) NEWCOPY

CEMT SET PROGRAM(MFRNSET) NEWCOPY

How to verify: Both display NORMAL with positive byte lengths.

3.Launch MYTR and Test F2:1 min.

Clear screen (**Ctrl + R**), type **MYTR**, and press Enter.

How to verify: Your green cursor block will land inside the **ENTER YEAR** field, allowing you to type data and press F2!

RERAN THE CICS PROGRAM JCL FOR BOTH HLQ D62WW AND TSOPS01 LOAD LIBRARIES.

THAT FIXED IT!

eRROR:The message Error Starting Browse Session.

The message **Error Starting Browse Session.**

FILE IS EMPTY

REQID = 0

IF REQID = 0

WOULD LIKE TO OPEN DATA ENTRY

```
TRANSACTION: MYTR PROGRAM: MFRNMAPP TASK: 0012313 APPLID: ES1ACICS DISPLAY: 00
STATUS:  COMMAND EXECUTION COMPLETE
EXEC CICS STARTBR
FILE ('SKILLS ')
RIDFLD ('2026')
REQID (0)
GTEQ
NOHANDLE
```

OFFSET:X'00100E'
RESPONSE: **NOTFND**

LINE:00348
EIBRESP=13

EIBFN=X'060C'

```
TRANSACTION: MYTR PROGRAM: MFRNMAPP TASK: 0012313 APPLID: ES1ACICS DISPLAY: 00
STATUS:  COMMAND EXECUTION COMPLETE
EXEC CICS STARTBR
  FILE ('SKILLS  ')
  RIDFLD ('2026')
  REQID (0)
  GTEQ
  NOHANDLE

OFFSET:X'00100E'      LINE:00348      EIBFN=X'060C'
RESPONSE: NOTFND      EIBRESP=13

ENTER:  CONTINUE
PF1 : UNDEFINED      PF2 : SWITCH HEX/CHAR      PF3 : END EDF SESSION
PF4 : SUPPRESS DISPLAYS  PF5 : WORKING STORAGE      PF6 : USER DISPLAY
PF7 : SCROLL BACK      PF8 : SCROLL FORWARD      PF9 : STOP CONDITIONS
PF10: PREVIOUS DISPLAY  PF11: EIB DISPLAY      PF12: ABEND USER TASK
```

Instead of letting **STARTBR** fail and lock the screen, you intercept **NOTFND** (**EIBRESP** = 13), populate your message area, set the map field attributes to unprotected, and re-issue **SEND MAP**.

JOB 2011 – GOOD